Galactic Dynamics & Star Clusters

An Overview Lecture for MSc and PhD Students

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Lecture Outline

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21. Chandrasekhar's Dynamical Friction
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Literature

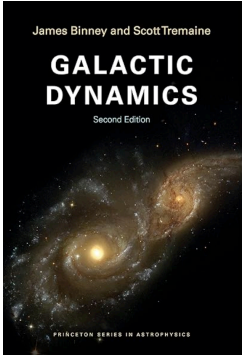
Literature

Books

- Galactic Dynamics, Binney & Tremaine 2008

Useful web pages:

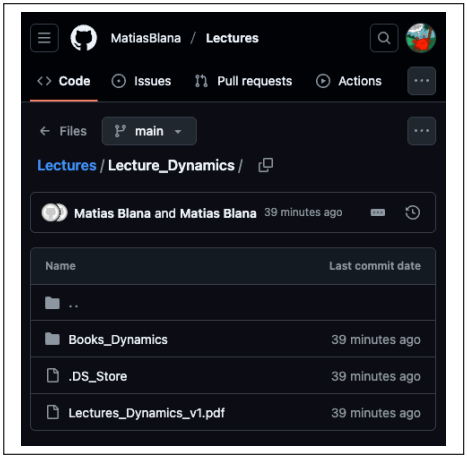
- Homepage: van den Bosch https://campuspress.yale.edu/vdbosch/teaching/
- Homepage: J.Bovy https://galaxiesbook.org/



Lecture Material

Repository:

- Slides in PDF
- Scripts (Python)



Available in GitHub:

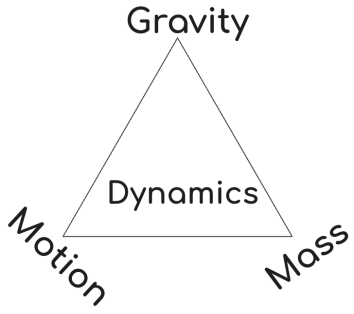
https://github.com/MatiasBlana/Lectures/tree/main/Lecture_Dynamics

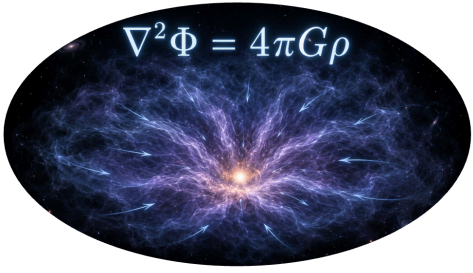
Introduction

Gravitational Dynamics

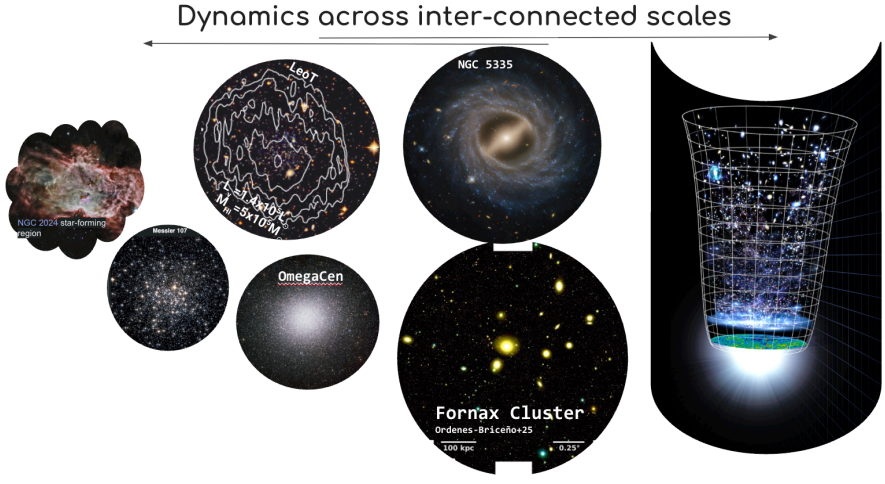
Gravitational dynamics studies the structure (x, ρ), motion ($\dot{x}$), and evolution ($\ddot{x}$) of astrophysical systems governed by gravity:

$$\frac{d^2\mathbf{x}}{dt^2} = -\nabla\Phi(\mathbf{x}, t) \tag{1}$$





Dynamics Across Astrophysical Scales



The same gravitational laws describe systems spanning many orders of magnitude in mass, size, and dynamical timescale.

Gravitational Dynamics

Main subfields, ordered by scale:

- **Celestial mechanics:** planets, satellites(arti.), few-body systems, and orbital resonances.
- **Stellar dynamics:** star clusters, stellar populations, and gravitational encounters.
- **Galactic dynamics:** galaxies, dark matter halos, orbital structure, and evolution.
- **Cosmological dynamics:** gravitational collapse, halo formation, and large-scale structure.

Common foundations: Newtonian gravity, gravitational potentials, the N-body problem, conservation laws, and dynamical timescales.

What is Stellar Dynamics?

Stellar dynamics studies the collective gravitational behavior of stars, from binaries and multiple stellar systems, to dense star clusters, and galactic stellar populations.

Main topics:

- The gravitational N-body problem.
- Collisionless and collisional dynamical regimes.
- Two-body relaxation, scattering, and evaporation.
- Mass segregation, core collapse, and instabilities.
- Dynamical evolution of star clusters within galaxies.

Key questions:

- How do stellar systems reach dynamical equilibrium?
- How do gravitational encounters redistribute energy?
- How do internal dynamics and galactic tides determine the survival and evolution of star clusters?

What is Galactic Dynamics?

Galactic dynamics studies the structure, kinematics, and evolution of galaxies and their constituent components under gravitational forces. It overlaps with stellar dynamics.

Main topics:

- Gravitational potentials and mass distributions: disks, bulges, and **dark matter** halos.
- Stellar and satellite orbits in galactic potentials.
- Collisionless dynamics: Boltzmann and Jeans equations.
- Gravitational instabilities, bars, spirals, and mergers.
- Tidal interactions, stripping, and secular evolution.

Key questions:

- How is mass distributed within galaxies?
- What determines their orbital structure and kinematics?
- How do galaxies evolve under internal and external perturbations?

Why do we study gravitational dynamics?

Dynamics allows us to infer what we cannot directly observe:

- Reconstruct mass distributions, including dark matter, from observed motions.
- Recover orbital histories and past gravitational interactions.
- Identify dynamical signatures: streams, tidal tails, velocity anisotropies, and kinematic substructures.

Dynamics allows us to predict how systems are formed and evolve:

- Gravitational instabilities, collapse, and structure formation.
- Energy and angular-momentum redistribution through encounters, resonances, and secular evolution.
- Mass loss, tidal disruption, mergers, and orbital decay.
- Long-term survival and dissolution of stellar systems.

Central challenge: Connect present-day observables to the **past evolution** and **future fate** of gravitational systems.

Star Clusters vs. Galaxies: Two Dynamical Regimes

Collisionless regime (most galaxies):

- Very large particle numbers: $N \gg 1$.
- $t_{\rm relax} \gg t_{\rm age}$.
- Evolution governed by the smooth gravitational potential.
- Described by the collisionless Boltzmann and Jeans equations.

Collisional regime (many star clusters):

- Smaller particle numbers and shorter relaxation times.
- $t_{\rm relax} \lesssim t_{\rm age}$.
- Two-body encounters redistribute energy and angular momentum.
- Relaxation, mass segregation, core collapse, and evaporation.

Key distinction: The relevant dynamical description depends on the **relaxation timescale**, not simply the physical size or density of the system.

From Individual Orbits to Collective Dynamics

Gravitational systems can be studied using two **complementary approaches**:

1. Individual orbital dynamics

- Equations of motion in gravitational potentials.
- Orbital invariants: energy and angular momentum.
- Orbital families, resonances, and perturbations.
- Numerical orbit integration.

2. Collective and statistical dynamics

- Phase-space distribution function $f(\mathbf{x}, \mathbf{v}, t)$.
- Collisionless Boltzmann and Jeans equations.
- Gravitational instabilities and collective evolution.
- Two-body relaxation and scattering when relevant.

Connecting both approaches: The gravitational potential $\Phi(\mathbf{x}, t)$ and its evolution.

Star Clusters as Dynamical Laboratories

Star clusters connect **stellar dynamics** with **galactic orbital dynamics**.

1. Self-gravitating systems

- Observable internal structure and kinematics.
- Evolution through gravitational encounters.
- Testbeds for relaxation, equilibrium, and dynamical evolution.

2. Galactic tracers

- Orbit within the host galaxy's potential.
- Respond to external tidal forces.
- Trace galactic mass distributions and assembly histories.

Connecting observations and models:

- Gaia/HST astrometry, spectroscopy, and photometry.
- Jeans modeling, orbital integration, and N-body simulations.

Central question: *How do internal dynamics and orbital histories within galaxies determine the structure, evolution, and survival of star clusters?*

Newtonian Gravity and Motion



Reputed descendants of Newton's apple tree at Trinity College, Cambridge (Wiki)

Newton's Universal Law of Gravitation (1687)

Isaac Newton, *Philosophiae Naturalis Principia Mathematica* (1687).

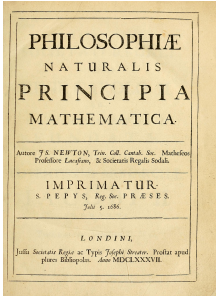
Gravitational force:

$$F = G\frac{m_1 m_2}{r^2} \tag{2}$$

Vector form:

$$\boldsymbol{F}_{12} = -Gm_1 m_2 \frac{\boldsymbol{r}_1 - \boldsymbol{r}_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|^3} \tag{3}$$

Newton's third law: $\boldsymbol{F}_{21} = -\boldsymbol{F}_{12}$.


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.
Auctore ꝗ. NEWTON, Trin. Coll. Cantab. Soc. Mathefeos Profeſſore Lucaſiano, & Societatis Regalis Sodali.
IMPRIMATUR.
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.
LONDINI,
Juſſu Societatis Regiæ ac Typis Joſephi Streater. Proſtat apud plures Bibliopolas. Anno MDCLXXXVII.

Principia (1687)

Gravitational constant:

$$\begin{aligned} G &= 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} \\ &\simeq 4.302 \times 10^{-6}\ \mathrm{kpc\,M_\odot^{-1}\,(km/s)^2} \\ &\simeq 4\pi^2\ \mathrm{AU^3\,M_\odot^{-1}\,yr^{-2}}\ \left(\text{Kepler's third law}: G = 4\pi^2\ \frac{a^3}{M\ P^2}\right) \end{aligned} \tag{4}$$

From Pairwise Forces to N-Body Dynamics

Gravitational acceleration of particle 1 due to particle 2:

$$\boldsymbol{a}_1 = \frac{\boldsymbol{F}_{12}}{m_1} = -Gm_2 \frac{\boldsymbol{r}_1 - \boldsymbol{r}_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|^3} \tag{5}$$

Superposition: for N gravitating particles,

$$\boxed{\frac{d^2\boldsymbol{r}_i}{dt^2} = -G \sum_{\substack{j=1 \\ j\neq i}}^{N} m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|^3}} \tag{6}$$

Key properties:

- Gravity is attractive and obeys superposition.
- Acceleration is **independent** of the test mass m_i.
- All particles interact gravitationally.
- For $N > 2$, general solutions require numerical methods.

Gravitational acceleration field

Similarly, the gravitational (acceleration) field vector form (on m_1 due to m_2):

$$\boldsymbol{g}_2 \;=\; -\,G\,\frac{m_2}{r^2}\,\hat{\boldsymbol{r}}_{12} \tag{7}$$

and on m_2 due to m_1):

$$\boldsymbol{g}_1 \;=\; -\,G\,\frac{m_1}{r^2}\,\hat{\boldsymbol{r}}_{21} \tag{8}$$

Where we know from Newton's third law:

$$m_1\boldsymbol{g}_2 \;=\; -m_2\boldsymbol{g}_1 \tag{9}$$

Acceleration, Momentum, Forces and effective/emergent Forces (*)

Momentum:

$$\mathbf{p} = m\,\mathbf{v} \tag{10}$$

Newton's Second Law:

$$\frac{d\mathbf{p}}{dt} = m\mathbf{a} = \mathbf{F} \tag{11}$$

Constant (inertial) mass:

$$\mathbf{a} = \frac{1}{m}\,\mathbf{F} \tag{12}$$

Gravity:

$$\mathbf{F}_{\rm g} = m_{\rm g}\,\boldsymbol{g} \tag{13}$$

If $m = m_{\rm g}$ (Equivalence Principle):

$$\mathbf{a} = \boldsymbol{g}\frac{\cancel{m_{\rm g}}}{\cancel{m}} \tag{14}$$

Lorentz Force (EM) [$m_p \sim 1.6\ 10^{-26}$ g $= 1836\ m_{e^+}$]:

$$\mathbf{a}_{\rm Lorentz} = \frac{\mathbf{F}_{\rm Lorentz}}{m} = \frac{q}{m}\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{15}$$

Ram Pressure (*):

$$\mathbf{a}_{\rm RP} = \frac{\mathbf{F}_{\rm RP}}{m} = \frac{-A\,\rho_{\rm env}}{m}\,v^2\,\widehat{\mathbf{v}} \tag{16}$$

Radiation Pressure (*):

$$\mathbf{a}_{\rm rad} = \frac{\mathbf{F}_{rad}}{m} = (1+\eta)\frac{I_{rad}\,Area}{m}\widehat{\mathbf{r}} \tag{17}$$

Dynamical Friction (*):

$$\mathbf{a}_{\rm Dyn} \propto -m\,\rho_{\rm env}\,v^{-2}\,\widehat{\mathbf{v}} \tag{18}$$

Gravitational Potential and Superposition

The gravitational potential per unit mass Φ defines the acceleration field:

$$\boldsymbol{g}(\boldsymbol{r}) = -\nabla\Phi(\boldsymbol{r}) \tag{19}$$

Contribution from a mass element at position $\boldsymbol{r}'$:

$$dM = \rho(\boldsymbol{r}')\, d^3r', \qquad d\Phi = -G\frac{dM}{|\boldsymbol{r}-\boldsymbol{r}'|} \tag{20}$$

Superposition: integrate over the entire mass distribution:

$$\boxed{\Phi(\boldsymbol{r}) = -G\int_{\mathbb{R}^3} \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\, d^3r'} \tag{21}$$

For discrete point masses:

$$\Phi(\boldsymbol{r}) = -G\sum_{i=1}^{N} \frac{m_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} \tag{22}$$

For a single point mass M at the origin:

From Field to Poisson's Equation

Start from (19) and take divergence:

$$\nabla \cdot \boldsymbol{g} \;=\; \nabla \cdot (-\nabla \Phi) \;=\; -\nabla^2 \Phi \tag{24}$$

Gauss's law for gravity (Newtonian) gives

$$\nabla \cdot \boldsymbol{g} \;=\; -4\pi G \rho \tag{25}$$

Equating (24) and (25) we get Poisson's equation:

$$\nabla^2 \Phi \;=\; 4\pi G \rho \tag{26}$$

We also have that the sum of densities (ρ_i) is the linear sum of their potentials (Φ_i):

$$\sum_i \nabla^2 \Phi_i = \nabla^2 \Phi_{\mathrm{tot}} = \sum_i 4\pi G \rho_i = 4\pi G \rho_{\mathrm{tot}} \tag{27}$$

In vacuum ($\rho = 0$) we get Laplace's equation:

$$\nabla^2 \Phi \;=\; 0 \tag{28}$$

Spherical Symmetry: Integrating Poisson

Spherical symmetry $\Rightarrow \Phi = \Phi(r)$:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\Phi}{dr}\right) \;=\; 4\pi G\rho(r) \tag{29}$$

Integrate once from 0 to r:

$$r^2\frac{d\Phi}{dr} \;=\; G\left[4\pi\int_0^r \rho(r')\,r'^2\,dr'\right] \;\equiv\; G\,M(r) \tag{30}$$

Hence

$$\frac{d\Phi}{dr} \;=\; \frac{GM(r)}{r^2}, \qquad M(r) \;=\; 4\pi\int_0^r \rho(r')\,r'^2\,dr' \tag{31}$$

Newton's Shell Theorem and Spherical Symmetry

For any spherically symmetric mass distribution $\rho(r)$, a spherical shell of matter exerts *no net gravitational force* on a test particle **inside** the shell. Only the mass **enclosed** within radius r contributes to the force as:

$$\mathbf{g}(r) = -\frac{G\,M(<r)}{r^2}\,\hat{\mathbf{r}}. \tag{32}$$

Implications:

- Spherical systems behave like a point mass for r outside the system.
- Internal dynamics depend only on $M(<r)$.
- Greatly simplifies Jeans modelling and orbit calculations. Many astrophysical spherical profiles: NFW, Burkert, etc.

But this symmetry is special:

- **Flattened (oblate/prolate) systems** $\rightarrow$ no shell cancellation.
- **Disks** have strong contributions from mass at all radii.
- Gravitational force must be computed via 2D or 3D integrals, or numerically, not from enclosed mass alone.

Solving Poisson's Equation: Overview

We seek the potential generated by a mass density:

$$\nabla^2 \Phi(\mathbf{x}) = 4\pi G \rho(\mathbf{x}) \tag{33}$$

Analytic solutions only exist for simple symmetries:

- Spherical systems
- Razor-thin axisymmetric disks
- Known analytic models (Plummer, Hernquist, MN, NFW)

General galaxies $\Rightarrow$ numerical solvers.

Boundary conditions matter:

- **Periodic** (cosmological simulations, FFT-based PM codes)
- **Isolated** (galaxies, star clusters)

Different numerical strategies exist for different geometries and resolutions.

FFT-Based Solvers (Particle–Mesh Methods)

FFT solvers exploit the Fourier-space representation:

$$\widehat{\Phi}(\mathbf{k}) \;=\; -\,\frac{4\pi G}{k^2}\,\widehat{\rho}(\mathbf{k})$$

Advantages:

- Fast: $\mathcal{O}(N \log N)$
- Excellent for large uniform grids
- Natural in periodic domains (cosmological boxes)

Limitations:

- Limited to grid resolution (force softened at cell size)

Used in:

- PM, P3M, TreePM codes (e.g. Gadget, Enzo, RAMSES Hybrid)
- Cosmological N-body simulations

Relaxation and Multigrid Solvers

Iterative relaxation methods:

- Jacobi, Gauss–Seidel, SOR (Successive Over-Relaxation)
- Update solution locally until convergence
- Robust but slow: $\mathcal{O}(N^{3/2})$

Multigrid (MG):

- Combines fine and coarse grids for optimal convergence
- Speed: $\mathcal{O}(N)$ — fastest known partial differential equation (PDE) solver

Used in:

- AMR codes: RAMSES, Athena, FLASH
- High-resolution galaxy simulations

Tree, TreePM and Method Comparison

Tree solvers:

- Hierarchical domain decomposition
- Distant masses approximated by multipole expansion
- Complexity: $\mathcal{O}(N \log N)$

TreePM hybrids:

- Short-range forces from Tree
- Long-range forces from FFT (PM)
- Used in modern large-volume cosmological codes

Gravitational Field from the Potential

$$\boldsymbol{g} = -\nabla\Phi \tag{34}$$

Cartesian (x, y, z):

$$\boldsymbol{g} = -\left(\partial_x\Phi,\ \partial_y\Phi,\ \partial_z\Phi\right) \tag{35}$$

Cylindrical (R, ϕ, z):

$$\boldsymbol{g} = -\left(\partial_R\Phi,\ \frac{1}{R}\partial_\phi\Phi,\ \partial_z\Phi\right) \tag{36}$$

Spherical (r, θ, ϕ):

$$\boldsymbol{g} = -\left(\partial_r\Phi,\ \frac{1}{r}\partial_\theta\Phi,\ \frac{1}{r\sin\theta}\partial_\phi\Phi\right) \tag{37}$$

Symmetries simplify the field:

- Spherical: $\Phi = \Phi(r) \Rightarrow$ only g_r.
- Axisymmetric: $\partial_\phi\Phi = 0 \Rightarrow g_\phi = 0$.

Spherical Harmonics in Gravitational Modeling

A general 3D gravitational potential can be expanded in spherical harmonics:

$$\Phi(r,\theta,\phi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} \Phi_{\ell m}(r)\, Y_{\ell m}(\theta,\phi) \tag{38}$$

- $\ell = 0$: spherical component (monopole).
- $\ell = 2$: quadrupole, flattening or triaxiality.
- $m \neq 0$: non-axisymmetric structure.

Applications in Galactic Dynamics:

M31 – Blaña et al. (2018)

- Made-to-Measure (NMAGIC).
- Spherical harmonics for dark matter.
- Expansion through order 16.

MNRAS – arXiv:1808.07494

Milky Way – Portail et al. (2017)

- Made-to-Measure (NMAGIC).
- Spherical harmonics for dark matter.
- Expansion through order 8.

MNRAS – arXiv:1608.07954

Hybrid potential solver: Spherical harmonics (halo) + cylindrical Fourier mesh (disk/bar).

Next application: Earth's gravitational potential (EGM2008).

Example: Earth's Gravitational Potential: Spherical Harmonics

The Earth's external gravitational potential can be expanded in **spherical harmonics**:

$$\Phi(r,\theta,\phi) = -\frac{GM_\oplus}{r}\left[1 + \sum_{\ell=2}^{L}\left(\frac{R_\oplus}{r}\right)^{\ell}\sum_{m=0}^{\ell}\bar{P}_{\ell m}(\cos\theta) \times \left(\bar{C}_{\ell m}\cos m\phi + \bar{S}_{\ell m}\sin m\phi\right)\right] \tag{39}$$

- ℓ, m: harmonic degree and order.
- $\bar{P}_{\ell m}$: normalized associated Legendre functions.
- $\bar{C}_{\ell m}, \bar{S}_{\ell m}$: mass distribution coefficients.
- $m = 0$: axisymmetric; $m \neq 0$: non-axisymmetric.
- Valid outside Earth's mass distribution: $r \gtrsim R_\oplus$, where $\nabla^2\Phi = 0$.

EGM2008: complete to degree/order 2159, with additional terms up to $\ell = 2190$.

Reference: Pavlis et al. (2012), EGM2008, JGR.

Earth's Gravitational Potential: J_2 Approximation

Retaining the dominant quadrupole term ($\ell = 2$, $m = 0$):

$$\Phi(r,\theta) = -\frac{GM_\oplus}{r}\left[1 - J_2\left(\frac{R_\oplus}{r}\right)^2 \times P_2(\cos\theta)\right] \qquad (40)$$

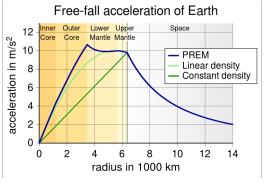
$$P_2(\cos\theta) = \frac{3\cos^2\theta - 1}{2} \qquad (41)$$

$J_2 \simeq 1.08263 \times 10^{-3}$

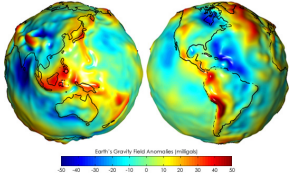
Physical interpretation:

- Earth's oblateness produces the quadrupole term.
- Axisymmetric: $\partial_\phi \Phi = 0$.
- Energy E and L_z are conserved.
- Orbital precession: nodes and pericenter.

But! Earth's potential ...



Radial gravity profile PREM (Preliminary Reference Earth Model Developed Dziewonski & Anderson 1981).



Gravity anomalies (mGal) GRACE (Gravity Recovery and Climate Experiment)

Collision-less Boltzmann Equation (CBE): connecting gravity and movement

Distribution Function and Equations of Motion

We can have a system of particles defined by the distribution function density $f(\boldsymbol{x}, \boldsymbol{v}, t)$ in a phase-space volume $d^3x\,d^3v$, with a number density of $dN = f\,d^3x\,d^3v$.

Equations of motion under $\Phi(\boldsymbol{x}, t)$:

$$\dot{\boldsymbol{x}} = \boldsymbol{v} \tag{42}$$

$$\dot{\boldsymbol{v}} = \boldsymbol{a} = -\nabla\Phi(\boldsymbol{x}, t) \tag{43}$$

Expanding the Lagrangian Derivative

Total (convective) derivative in phase space:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \dot{\boldsymbol{x}} \cdot \nabla_x f + \dot{\boldsymbol{v}} \cdot \nabla_v f \tag{44}$$

Insert (42)–(43):

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla_x f - (\nabla \Phi) \cdot \nabla_v f \tag{45}$$

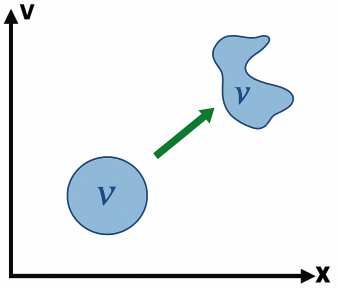
Collisionless Boltzmann Equation (Vlasov)

Liouville's theorem (phase-space incompressibility for Hamiltonian flow):

$$\frac{Df}{Dt} = 0 \iff \frac{\partial f}{\partial t} + \boldsymbol{v}\cdot\nabla_x f - (\nabla\Phi)\cdot\nabla_v f = 0 \tag{46}$$

This means f is constant along characteristics $(\boldsymbol{x}(t), \boldsymbol{v}(t))$.

Note*: collisional systems we have that $\frac{Df}{Dt} = I(t, x, v)$, called the **Fokker–Planck equation**, very hard to solve, but useful for some cases.



CBE: Zeroth Moment (Continuity)

Definitions of Velocity Moments

$$\rho(\boldsymbol{x}, t) \;=\; \int f(\boldsymbol{x}, \boldsymbol{v}, t)\, d^3v \tag{47}$$

$$\rho\, u_i(\boldsymbol{x}, t) \;=\; \int v_i\, f\, d^3v \tag{48}$$

$$\rho\, \langle v_i v_j \rangle(\boldsymbol{x}, t) \;=\; \int v_i v_j\, f\, d^3v \tag{49}$$

Velocity dispersion tensor:

$$\sigma_{ij}^2(\boldsymbol{x}, t) \;\equiv\; \langle (v_i - u_i)(v_j - u_j) \rangle \;=\; \langle v_i v_j \rangle - u_i u_j \tag{50}$$

For the simplest case, a collection of N particles:

$$\sigma_i^2(\boldsymbol{x}, t) \;=\; \langle v_i^2 \rangle - u_i^2 \tag{51}$$

Integrate CBE over d^3v

Integrate (46) over velocities and use that $\int \nabla_v \cdot (\cdots)\, d^3v = 0$ if $f \to 0$ sufficiently fast:

$$\int \frac{\partial f}{\partial t} d^3v + \int v_j \frac{\partial f}{\partial x_j} d^3v \;=\; 0 \tag{52}$$

Switch derivatives and integrals:

$$\frac{\partial}{\partial t}\left(\int f d^3v \right) + \frac{\partial}{\partial x_j}\left(\int v_j f d^3v \right) \;=\; 0 \tag{53}$$

Hence the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) \;=\; 0 \tag{54}$$

CBE: First Moment (Jeans)

Multiply CBE by v_i and Integrate

Start from (46), multiply by v_i and integrate over d^3v:

$$\frac{\partial}{\partial t}\left(\int v_i f d^3v\right) + \frac{\partial}{\partial x_j}\left(\int v_i v_j f d^3v\right) + \int v_i \left[-\partial_k \Phi \frac{\partial f}{\partial v_k}\right] d^3v = 0 \tag{55}$$

The last term is integrated by parts in v_k:

$$\int v_i \left(-\partial_k \Phi\right) \frac{\partial f}{\partial v_k} d^3v = -\partial_k \Phi \int v_i \frac{\partial f}{\partial v_k} d^3v \tag{56}$$

Evaluating the Velocity-Space Surface Term

Use $\int v_i \partial f/\partial v_k \, d^3v = \left[v_i f\right]_{v_k=-\infty}^{+\infty} - \int \delta_{ik} f \, d^3v$. If $v_i f \to 0$ at $|v_k| \to \infty$:

$$\int v_i \frac{\partial f}{\partial v_k} \, d^3v \;=\; -\int \delta_{ik} f \, d^3v \;=\; -\delta_{ik}\,\rho \tag{57}$$

Therefore the gravity term in (55) is

$$-\,\partial_k \Phi \left(-\,\delta_{ik}\rho\right) \;=\; +\,\rho\,\partial_i \Phi \tag{58}$$

Putting terms together:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho \langle v_i v_j \rangle)}{\partial x_j} + \rho\,\partial_i \Phi \;=\; 0 \tag{59}$$

Convective (Jeans) Form and Its Analogy to Euler's Equation

Velocity–moment split:

$$\langle v_i v_j \rangle \;=\; u_i u_j \;+\; \sigma_{ij}^2 \tag{60}$$

Insert (60) into the first-moment equation (59) and use the continuity equation (54) to obtain:

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = - \frac{\partial (\rho \, \sigma_{ij}^2)}{\partial x_j} - \; \rho \, \partial_i \Phi. \tag{61}$$

This is the **Euler-like Jeans momentum equation** for a collisionless stellar fluid, where the tensor $\rho \sigma_{ij}^2$ plays the role of a *pressure/anisotropy stress*.

Comparison with the collisional Euler equation (with gravity):

$$\rho\left(\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}\right)=-\partial_i P-\rho\,\partial_i\Phi. \tag{62}$$

Analogy:

$$\boxed{\frac{\partial(\rho\,\sigma_{ij}^2)}{\partial x_j}\;\longleftrightarrow\;\partial_i P}$$

Collisionless systems: anisotropic pressure tensor $\rho\sigma_{ij}^2$ vs. collisional gases: scalar pressure P.

EXTRA: CBE: Second Moment (Dispersion Evolution)

Multiply CBE by $v_i v_j$ and Integrate

Multiply (46) by $v_i v_j$ and integrate:

$$\frac{\partial(\rho\langle v_i v_j\rangle)}{\partial t} + \frac{\partial(\rho\langle v_i v_j v_k\rangle)}{\partial x_k} + \rho\left(u_i\partial_j\Phi + u_j\partial_i\Phi\right) = 0 \tag{63}$$

Sketch of gravity term: Integrate by parts twice in v_k, keeping track that $\int v_i v_j\,\partial f/\partial v_k\,d^3v = -\delta_{ik}\int v_j f\,d^3v - \delta_{jk}\int v_i f\,d^3v$, leading to (63).

Central Moments and the Q_{ijk} Flux

Decompose the third raw moment:

$$\langle v_i v_j v_k \rangle = u_i u_j u_k + u_i \sigma_{jk}^2 + u_j \sigma_{ik}^2 + u_k \sigma_{ij}^2 + Q_{ijk} \tag{64}$$

where the third central moment (heat/dispersion flux) is

$$Q_{ijk} \equiv \big\langle (v_i - u_i)(v_j - u_j)(v_k - u_k) \big\rangle \tag{65}$$

Inserting (64) into (63) and using (54), (61) yields:

$$\rho \left(\frac{\partial \sigma_{ij}^2}{\partial t} + u_k \frac{\partial \sigma_{ij}^2}{\partial x_k} + \sigma_{ik}^2 \frac{\partial u_j}{\partial x_k} + \sigma_{jk}^2 \frac{\partial u_i}{\partial x_k} \right) + \frac{\partial (\rho Q_{ijk})}{\partial x_k} = -\rho \left(u_i \partial_j \Phi + u_j \partial_i \Phi \right) + (\text{terms canceled by continuity/Jeans}) \tag{66}$$

Closure and Physical Meaning

- The **moment hierarchy** is unclosed: Eq. (66) depends on Q_{ijk}.
- **Closures** (e.g., $Q_{ijk}=0$, or gradient-diffusion models) are needed for practical modeling.
- In steady, symmetric systems, many terms simplify, leading to classic Jeans models of galaxies and clusters.

Appendix: Circular Velocity Link (Preview)

Centripetal Balance $\Rightarrow v_c$–Φ–$M(<r)$

Centripetal acceleration:

$$a_{\rm cent} = \frac{v_c^2}{r} \tag{67}$$

Gravitational acceleration (spherical):

$$a_r = -\frac{d\Phi}{dr} = -\frac{GM(<r)}{r^2} \tag{68}$$

Circular orbit: set $a_{\rm cent} = -a_r$:

$$v_c^2(r) = r\frac{d\Phi}{dr} \tag{69}$$

$$v_c^2(r) = \frac{GM(<r)}{r} \tag{70}$$

Summary of Part I

Key Takeaways

- Newtonian gravity: $\boldsymbol{a} = -\nabla\Phi$, with $\nabla^2\Phi = 4\pi G\rho$.
- CBE: $\dfrac{Df}{Dt} = 0$ encodes phase-space conservation.
- Moments $\Rightarrow$ continuity (54), Jeans (61), and dispersion evolution (66).
- The hierarchy requires **closure** (assumptions on higher moments).

Jeans: Spherical Systems

From CBE to Spherical Jeans

Start from the steady, spherical CBE first moment (no $\partial/\partial t$, isotropy not yet assumed). Let $\rho(r)$ be the tracer density, $\overline{v_r^2}(r)$ and $\overline{v_\theta^2}(r) = \overline{v_\phi^2}(r)$ the moments. The radial Jeans equation is

$$\frac{d(\rho\,\overline{v_r^2})}{dr} + \frac{2\rho}{r}\left(\overline{v_r^2} - \overline{v_\theta^2}\right) = -\,\rho\,\frac{d\Phi}{dr}. \tag{71}$$

Define anisotropy

$$\beta(r) \;\equiv\; 1 - \frac{\overline{v_\theta^2}}{\overline{v_r^2}}. \tag{72}$$

Divide (71) by ρ to obtain

$$\frac{1}{\rho}\frac{d(\rho\,\overline{v_r^2})}{dr} + \frac{2\beta\,\overline{v_r^2}}{r} = -\,\frac{d\Phi}{dr}. \tag{73}$$

Useful Integral Solution Form

Using (31) ($d\Phi/dr = GM(r)/r^2$) in (73):

$$\frac{d(\rho\,\overline{v_r^2})}{dr} + \frac{2\beta\,\rho\,\overline{v_r^2}}{r} = -\,\rho\,\frac{GM(r)}{r^2}. \tag{74}$$

This linear first-order ODE admits the integrating factor $I(r) = \exp\left(\int \frac{2\beta}{r}\,dr\right)$, giving

$$\rho\,\overline{v_r^2}(r)\,I(r) = \int_r^\infty \rho(s)\,\frac{GM(s)}{s^2}\,I(s)\,ds, \quad \Rightarrow \quad \overline{v_r^2}(r) = \frac{1}{\rho(r)I(r)}\int_r^\infty \rho(s)\,\frac{GM(s)}{s^2}\,I(s)\,ds. \tag{75}$$

For isotropy ($\beta = 0$), $I \equiv 1$.

Isotropic Case and Isothermal Sphere

Isotropy: $\beta = 0 \Rightarrow \overline{v_\theta^2} = \overline{v_r^2}$. Jeans becomes

$$\frac{d(\rho\,\overline{v_r^2})}{dr} = -\,\rho\,\frac{d\Phi}{dr}. \tag{76}$$

Singular isothermal sphere with constant 1D dispersion σ:

$$\rho(r) = \frac{\sigma^2}{2\pi G r^2}, \qquad \frac{d\Phi}{dr} = \frac{2\sigma^2}{r}. \tag{77}$$

Then the circular velocity is

$$v_c^2(r) = r\,\frac{d\Phi}{dr} = 2\sigma^2 = \text{const.} \tag{78}$$

EXTRA: Jeans: Axisymmetric Disks

Jeans in Cylindrical Coordinates (Steady, Axisymmetric)

Assume $\partial/\partial t = 0$ and $\partial/\partial \phi = 0$. The radial and vertical Jeans equations are:

$$\frac{\partial(\rho\overline{v_R^2})}{\partial R} + \frac{\partial(\rho\overline{v_R v_z})}{\partial z} + \rho\left(\frac{\overline{v_R^2} - \overline{v_\phi^2}}{R}\right) = -\rho\frac{\partial\Phi}{\partial R}, \tag{79}$$

$$\frac{\partial(\rho\overline{v_z^2})}{\partial z} + \frac{\partial(\rho\overline{v_R v_z})}{\partial R} + \frac{\rho\overline{v_R v_z}}{R} = -\rho\frac{\partial\Phi}{\partial z}. \tag{80}$$

These relate gradients of stress components to the gravitational force.

Thin Isothermal Disk: Vertical Structure (Full Steps)

Assume thin disk near mid-plane with negligible cross term $\overline{v_R v_z} \approx 0$ and $\overline{v_z^2} = \sigma_z^2 =$ const. From (80):

$$\frac{d(\rho\sigma_z^2)}{dz} = -\rho\,\frac{d\Phi}{dz}. \tag{81}$$

Combine with Poisson (local sheet):

$$\frac{d^2\Phi}{dz^2} = 4\pi G\rho. \tag{82}$$

Differentiate (81) and use (82):

$$\sigma_z^2\frac{d^2\rho}{dz^2} = -\rho\,\frac{d^2\Phi}{dz^2} - \frac{d\rho}{dz}\frac{d\Phi}{dz} = -4\pi G\rho^2 - \frac{d\rho}{dz}\frac{1}{\rho}\frac{d(\rho\sigma_z^2)}{dz}. \tag{83}$$

With σ_z const, an exact solution is the Spitzer sech^2 profile:

$$\rho(z) = \rho_0\,\mathrm{sech}^2\left(\frac{z}{z_0}\right), \qquad z_0 = \frac{\sigma_z}{\sqrt{2\pi G\rho_0}}. \tag{84}$$

Radial Force Balance and Asymmetric Drift

From (79) (neglect ∂_z term for an infinitely thin disk in mid-plane):

$$\frac{1}{\rho}\frac{\partial(\rho\overline{v_R^2})}{\partial R} + \frac{\overline{v_R^2} - \overline{v_\phi^2}}{R} = -\frac{\partial \Phi}{\partial R}. \tag{85}$$

Define the circular velocity via $v_c^2(R) \equiv R\partial_R\Phi$. Then

$$\overline{v_\phi^2} = v_c^2 - \overline{v_R^2}\left[\frac{\partial \ln(\rho\overline{v_R^2})}{\partial \ln R} + 1\right]. \tag{86}$$

In terms of $\overline{v_\phi}$ (assuming small skewness),

$$v_c^2 - \overline{v_\phi}^2 \simeq \overline{v_R^2}\left[\frac{\partial \ln(\rho\overline{v_R^2})}{\partial \ln R} + 1\right]. \tag{87}$$

This lag ($\overline{v_\phi} < v_c$) is the *asymmetric drift* meaning that $v_c^2 \neq \overline{v_\phi}^2$.

Exponential Disk: Surface Density and Bessel-Function Formula

Infinitesimally thin exponential disk:

$$\Sigma(R) = \Sigma_0 \, \exp(-R/R_d). \tag{88}$$

Exact circular velocity (Freeman 1970) with $y = R/(2R_d)$:

$$v_c^2(R) = 4\pi G \Sigma_0 R_d \, y^2 \Big[I_0(y) K_0(y) - I_1(y) K_1(y) \Big], \tag{89}$$

where I_n and K_n are modified Bessel functions. The curve rises linearly at small R and peaks near $R \simeq 2.2 R_d$.

Including Dispersion: Asymmetric Drift for Disks

If the radial dispersion $\sigma_R(R)$ is non-zero, then $v_c^2 \neq \overline{v_\phi}^2$ i.e.,

$$v_c^2 - \overline{v_\phi}^2 = \sigma_R^2 \left[\frac{\partial \ln(\Sigma \sigma_R^2)}{\partial \ln R} + 1 \right]. \tag{90}$$

Thus $\overline{v_\phi}$ is lower where σ_R is large and/or declines steeply with R, especially in the inner disk.

Summary of Part II

Key Takeaways

- Spherical Jeans with anisotropy $\beta(r)$ provides $\overline{v_r^2}(r)$ once $\rho(r)$ and $M(r)$ are given.
- Axisymmetric Jeans connects stress gradients to $\partial_R \Phi$ and $\partial_z \Phi$; asymmetric drift quantifies $\overline{v_\phi} < v_c$.
- Exponential disks have closed-form $v_c(R)$ with modified Bessel functions; peak at $\simeq 2.2 R_d$.
- MN disk and NFW halo combine naturally to reproduce flat outer rotation curves.

Common Equilibrium Profiles

Plummer Sphere (Cored, when $r \to 0$ then $\rho \to$const)

Density:

$$\rho(r) = \frac{3M}{4\pi a^3}\left(1 + \frac{r^2}{a^2}\right)^{-5/2} \tag{91}$$

Enclosed mass:

$$M(r) = M\frac{r^3}{(r^2 + a^2)^{3/2}} \tag{92}$$

Potential:

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}} \tag{93}$$

Circular velocity:

$$v_c^2(r) = \frac{GMr^2}{(r^2 + a^2)^{3/2}} \tag{94}$$

Hernquist (Cuspy, when $r \to 0$ then $\rho \to \infty \approx$ SMBH?)

$$\rho(r) = \frac{M}{2\pi} \frac{a}{r(r+a)^3} \tag{95}$$

$$M(r) = M \frac{r^2}{(r+a)^2} \tag{96}$$

$$\Phi(r) = -\frac{GM}{r+a} \tag{97}$$

$$v_c^2(r) = \frac{GMr}{(r+a)^2} \tag{98}$$

Miyamoto–Nagai Disk (Axisymmetric)

Potential:

$$\Phi(R,z) = -\frac{GM}{\sqrt{R^2 + \left(a + \sqrt{z^2 + b^2}\right)^2}} \tag{99}$$

Density:

$$\rho(R,z) = \frac{b^2 M}{4\pi} \frac{aR^2 + (a + 3\sqrt{z^2 + b^2})(a + \sqrt{z^2 + b^2})^2}{\left[R^2 + (a + \sqrt{z^2 + b^2})^2\right]^{5/2} (z^2 + b^2)^{3/2}} \tag{100}$$

Mid-plane circular velocity:

$$v_c^2(R) = \frac{GMR^2}{\left[R^2 + (a+b)^2\right]^{3/2}} \tag{101}$$

NFW Halo (Cosmological)

$$\rho(r) = \frac{\rho_0}{(r/r_s)(1 + r/r_s)^2} \tag{102}$$

$$M(r) = 4\pi\rho_0 r_s^3 \left[\ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s}\right] \tag{103}$$

$$\Phi(r) = -4\pi G\rho_0 r_s^3 \frac{\ln(1 + r/r_s)}{r} \tag{104}$$

$$v_c^2(r) = \frac{GM(r)}{r} \tag{105}$$

Normalization with concentration $c = r_{200}/r_s$ and $\rho_{\rm crit}$:

$$\rho_0 = \frac{200}{3} \frac{c^3\, \rho_{\rm crit}}{\ln(1 + c) - c/(1 + c)}. \tag{106}$$

Profiles: Summary Table

Model	$\rho(r \to 0)$	$\rho(r \to \infty)$	$M(r)$	Notes
Plummer	const. core	r^{-5}	finite $\to M$	Cored, finite mass
Hernquist	r^{-1}	r^{-4}	finite $\to M$	Cuspy bulge-like
Miyamoto–Nagai	finite	r^{-3}	—	Flattened disk
NFW	r^{-1}	r^{-3}	log-divergent	Cosmological halo

Composite Rotation Curves (Plummer + MN + NFW)

Total Potential and Circular velocity

Assume total potential as a sum (as we have that $\nabla^2 \sum_i \Phi_i = \Phi_{\rm tot} = \sum_i \rho_i = \rho_{\rm tot}$):

$$\Phi_{\rm tot}(R,z) = \Phi_{\rm Plummer}(r) + \Phi_{\rm MN}(R,z) + \Phi_{\rm NFW}(r), \qquad r = \sqrt{R^2+z^2}. \tag{107}$$

Mid-plane circular velocity adds in quadrature of contributions:

$$v^2_{c,\rm tot}(R) = v^2_{c,\rm Plummer}(R) + v^2_{c,\rm MN}(R) + v^2_{c,\rm NFW}(R). \tag{108}$$

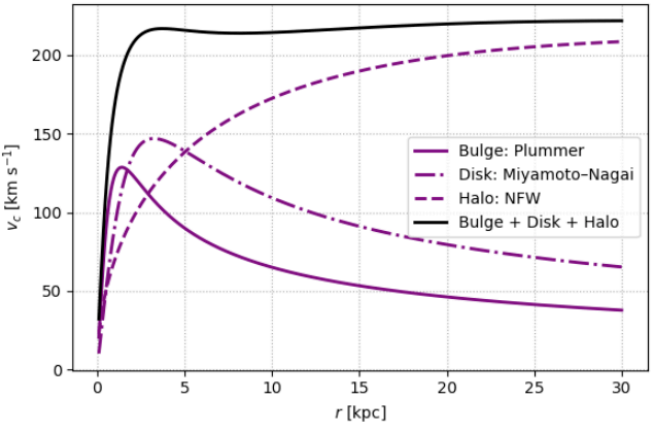
Where

$$v^2_c(R) = \frac{GMR^2}{(R^2+a^2)^{3/2}}, \tag{109}$$

$$v^2_{c,\rm MN}(R) = \frac{GM_{\rm d}R^2}{\left[R^2+(a+b)^2\right]^{3/2}}, \tag{110}$$

$$v^2_{c,\rm NFW}(R) = \frac{4\pi G\rho_0 r_s^3}{R}\left[\ln(1+R/r_s) - \frac{R/r_s}{1+R/r_s}\right]. \tag{111}$$

Figure: Rotation Curves (MN + NFW)



Circular Velocity, Potential Gradient, and Signs

Centripetal Balance and Signs

By definition,

$$a_{\rm cent} = \frac{v_c^2}{r}, \qquad a_r = -\frac{d\Phi}{dr}. \tag{112}$$

Circular orbit requires $a_{\rm cent} = -a_r$:

$$v_c^2(r) = r\frac{d\Phi}{dr}. \tag{113}$$

For spheres, $d\Phi/dr > 0$ (outward) and force is inward; in ring/disk centers, $d\Phi/dr$ can change sign, implying outward forces near the center.

Where v_c is Meaningful

- $v_c(r)$ is defined where gravity provides *centripetal* acceleration.
- In non-spherical potentials (e.g., triaxial halos), closed circular orbits may not exist at all radii; one uses $v_{\mathrm{circ}}(R,0) = \sqrt{R\partial_R \Phi(R,0)}$ as a diagnostic.
- Observed mean rotation $\overline{v_\phi}$ is *not* v_c in general; dispersion reduces $\overline{v_\phi}$ (asymmetric drift).

Virial Ratio and Equilibrium

Virial Theorem

Any gravitational system has a total energy that is the sum of the kinetic and potential energies

$$E = K + U. \tag{114}$$

For a bound isolated system we will have that

$$E = K + U \leq 0. \tag{115}$$

A system is in virial equilibrium if it is in a steady state (or after a long-time average of its moment of inertia $\langle \ddot{I} \rangle = 0$), where its total kinetic and potential energies obey the relation:

$$2K + U = 0. \tag{116}$$

Virial Ratios: Definitions and Meaning

Define

$$\alpha_{\rm vir} \equiv \frac{2K}{|U|}, \qquad Q \equiv \frac{K}{|U|} = \frac{1}{2}\alpha_{\rm vir}. \tag{117}$$

Virial equilibrium:

$$\alpha_{\rm vir} = 1 \quad \Longleftrightarrow \quad 2K = |U|. \tag{118}$$

Interpretation:

- $\alpha_{\rm vir} < 1$: sub-virial $\Rightarrow$ tendency to collapse/contract.
- $\alpha_{\rm vir} > 1$: super-virial $\Rightarrow$ tendency to expand/unbind.

Isotropic Systems: Link to Dispersion

For an isotropic system with 1D dispersion σ:

$$K = \frac{3}{2} M \sigma^2. \tag{119}$$

At equilibrium ($2K = -U$):

$$\sigma^2 = \frac{|U|}{3M}. \tag{120}$$

These relations underlie simple dynamical mass estimators.

Examples: Uniform Sphere & Plummer

Uniform-density sphere (M, R):

$$U = -\frac{3}{5}\frac{GM^2}{R}, \qquad \sigma^2 = \frac{GM}{5R}. \tag{121}$$

Plummer model (M, a):

$$U = -\frac{3\pi}{32}\frac{GM^2}{a}, \qquad \sigma^2 = \frac{\pi}{32}\frac{GM}{a}. \tag{122}$$

The latter is a *global* scaling; local $\sigma(r)$ follows from Jeans.

Observed Virial-Type Estimators

For pressure-supported systems (projected):

$$M \simeq k \frac{R_h \sigma_{\rm los}^2}{G}, \tag{123}$$

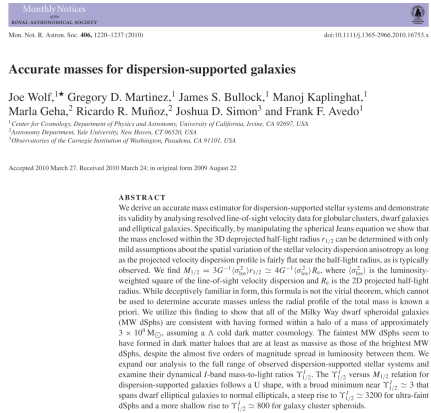
with R_h a characteristic projected radius (often R_e) and k a structure factor ($k \sim 3$ for many spheroids). If ordered rotation contributes, use the second moment

$$\langle v^2 \rangle \rightarrow \langle v_{\rm rot}^2 + \sigma^2 \rangle. \tag{124}$$

Caveat: k depends on geometry, anisotropy, and light–mass mismatch.

Wolf et al. (2010):

Half-Light Mass



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Accurate masses for dispersion-supported galaxies

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ABSTRACT

We derive an accurate mass estimator for dispersion-supported stellar systems and demonstrate its validity by analysing resolved line-of-sight velocity data for globular clusters, dwarf galaxies and elliptical galaxies. Specifically, by manipulating the spherical Jeans equation we show that the mass enclosed within the 3D deprojected half-light radius $r_{1/2}$ can be determined with only mild assumptions about the spatial variation of the stellar velocity dispersion anisotropy as long as the projected velocity dispersion profile is fairly flat near the half-light radius, as is typically observed. We find $M_{1/2} = 3G^{-1}\langle\sigma^2_{\rm los}\rangle r_{1/2} \simeq 4G^{-1}\langle\sigma^2_{\rm los}\rangle R_e$, where $\langle\sigma^2_{\rm los}\rangle$ is the luminosity-weighted square of the line-of-sight velocity dispersion and R_e is the 2D projected half-light radius. While deceptively familiar in form, this formula is not the virial theorem, which cannot be used to determine accurate masses unless the radial profile of the total mass is known a priori. We utilize this finding to show that all of the Milky Way dwarf spheroidal galaxies (MW dSphs) are consistent with having formed within a halo of a mass of approximately 3×10^7 $M_{\odot}$, assuming a Λ cold dark matter cosmology. The faintest MW dSphs seem to have formed in dark matter haloes that are at least as massive as those of the brightest MW dSphs, despite the almost five orders of magnitude spread in luminosity between them. We expand our analysis to the full range of observed dispersion-supported stellar systems and examine their dynamical I-band mass-to-light ratios $\Upsilon_{1/2}^I$. The $\Upsilon_{1/2}^I$ versus $M_{1/2}$ relation for dispersion-supported galaxies follows a U shape, with a broad minimum near $\Upsilon_{1/2}^I \simeq 3$ that spans dwarf elliptical galaxies to normal ellipticals, a steep rise to $\Upsilon_{1/2}^I \simeq 3200$ for ultra-faint dSphs and a more shallow rise to $\Upsilon_{1/2}^I \simeq 800$ for galaxy cluster spheroids.

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Core Result and Why It's Robust

For spherical, dispersion-supported systems with slowly varying $\sigma_{\rm los}(R)$, **Wolf et al. (2010)** showed the mass at the deprojected half-light radius $r_{1/2}$ is nearly anisotropy-independent:

$$M_{1/2} \equiv M(< r_{1/2}) = \frac{3}{G} \langle \sigma^2_{\rm los} \rangle \, r_{1/2}. \tag{125}$$

Using $r_{1/2} \simeq \frac{4}{3} R_e$ (for typical profiles):

$$M_{1/2} \simeq \frac{4}{G} \langle \sigma^2_{\rm los} \rangle \, R_e. \tag{126}$$

Sketch of the Anisotropy Insensitivity

Spherical Jeans (steady):

$$\frac{d(\nu\overline{v_r^2})}{dr} + \frac{2\beta\nu\overline{v_r^2}}{r} = -\nu\,\frac{GM(r)}{r^2}, \tag{127}$$

with tracer density $\nu(r)$ and anisotropy $\beta(r) = 1 - \overline{v_\theta^2}/\overline{v_r^2}$. The *projected* second moment $\langle\sigma_{\rm los}^2\rangle$ depends on (ν, β), but near the radius r_3 where the 3D slope satisfies

$$\left.\frac{d\ln\nu}{d\ln r}\right|_{r=r_3} = -3, \tag{128}$$

the β-dependence cancels to leading order, giving

$$M(r_3) \;=\; \frac{3}{G}\,\langle\sigma_{\rm los}^2\rangle\, r_3. \tag{129}$$

For many light profiles, $r_3 \approx r_{1/2}$, yielding (125).

What Dispersion to Use (Luminosity Weighting)

Define the luminosity-weighted LOS dispersion:

$$\langle \sigma_{\rm los}^2 \rangle \;=\; \frac{\displaystyle\int_0^\infty I(R)\,\sigma_{\rm los}^2(R)\,2\pi R\,dR}{\displaystyle\int_0^\infty I(R)\,2\pi R\,dR}, \tag{130}$$

where $I(R)$ is the surface brightness. If $\sigma_{\rm los}(R)$ is flat near R_e, a single global $\sigma_{\rm los}$ is a good proxy.

Numerical Forms (Handy Units)

With $G = 4.302 \times 10^{-3}$ pc $(\mathrm{km\ s^{-1}})^2\ M_\odot^{-1}$:

$$\frac{M_{1/2}}{M_\odot} \simeq 930 \left(\frac{\langle \sigma_{\rm los}^2 \rangle}{(\mathrm{km\,s^{-1}})^2}\right) \left(\frac{R_e}{\mathrm{pc}}\right), \tag{131}$$

$$\frac{M_{1/2}}{M_\odot} \simeq 700 \left(\frac{\langle \sigma_{\rm los}^2 \rangle}{(\mathrm{km\,s^{-1}})^2}\right) \left(\frac{r_{1/2}}{\mathrm{pc}}\right). \tag{132}$$

Assumptions and Scope

- Approximately spherical, dispersion-supported, equilibrium systems.
- $\sigma_{\rm los}(R)$ not strongly rising/falling near R_e.
- Insensitivity to β holds *near* $r_{1/2}$; away from it, anisotropy matters.
- For rotating systems, include ordered rotation in the second moment if needed: $\langle \sigma^2_{\rm los} \rangle \to \langle \sigma^2_{\rm los} + v^2_{\rm rot,los} \rangle$.

Worked Examples

Example: Dwarf Galaxy

Given:

$$R_e = 300\ \mathrm{pc}, \qquad \langle \sigma_{\mathrm{los}} \rangle = 8\ \mathrm{km\,s^{-1}}.$$

Using Eq. (131):

$$M_{1/2} \simeq 930 \times (8)^2 \times 300\ M_\odot = 1.8 \times 10^7\ M_\odot. \tag{133}$$

Mass-to-light inside $r_{1/2}$:

$$\Upsilon_{1/2} \equiv \frac{M_{1/2}}{L_{1/2}} \simeq \frac{2M_{1/2}}{L_{\mathrm{tot}}}. \tag{134}$$

Large $\Upsilon_{1/2}$ indicates dark-matter dominance.

Example: Ω Centauri (NGC 5139)

Adopt:

$$R_e \simeq 10\ \mathrm{pc}, \qquad \langle \sigma_{\mathrm{los}} \rangle \simeq 13\ \mathrm{km\,s^{-1}}.$$

Projected estimator (Eq. (131)):

$$M_{1/2} \ \simeq\ 930 \times 13^2 \times 10\ M_\odot \ \approx\ 1.57 \times 10^6\ M_\odot. \tag{135}$$

Notes:

- Ω Cen has modest rotation; include v_{rot} if using a second-moment estimator.
- Uncertainties in R_e and σ_{los} propagate linearly to $M_{1/2}$ in these forms.

When Virial/Wolf Estimators Can Mislead

- Non-equilibrium (mergers, rapid tides, gas expulsion) $\Rightarrow \langle \ddot{I} \rangle \neq 0$.
- Strong anisotropy gradients near R_e.
- Triaxiality or rotation-dominated kinematics without proper second-moment treatment.
- Mismatch between tracer light and mass distribution (e.g., strong M/L gradients, embedded dark core).

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nProFit: a tool for fitting the surface brightness profiles of star clusters with dynamical models

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ABSTRACT

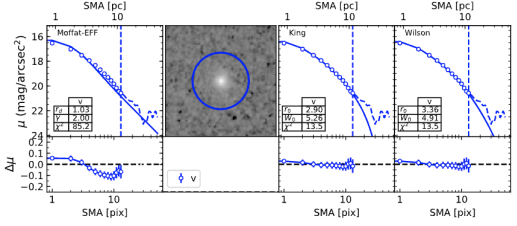


Figure 11. Dynamical models (Moffat-EFF, King and Wilson) fitting performed by nProFit to the surface brightness profile of a cluster in the mock sample generated with mksample (top panel). Fitting residuals (bottom panel). The fitting radius is shown with the vertical dashed line, the observed SBP with the empty dots and the fitted models with solid lines.

Figure: Applications of Jeans Equations:
https://ui.adsabs.harvard.edu/abs/2022arXiv220102881C/abstract

Dynamical evolution processes in star clusters

Internal Dynamical Processes in Star Clusters

Star clusters evolve through a set of **internal, self-regulated** dynamical mechanisms:

Two-body relaxation

- Gravitational encounters exchange energy and angular momentum.
- Pseudo-equilibria: drives an evolution of the initial energy distribution of the system.
- Sets t_{relax}, t_{evap}, and global evolution.

Scattering and evaporation

- High-energy encounters eject stars above escape velocity **(gravitational slingshot)**.
- Leads to long-term mass loss ($t_{\mathrm{evap}} \sim 100\, t_{\mathrm{relax}}$).

Two-body Scattering, Coulomb Logarithm, and Relaxation time

Context: In a stellar system, stars interact through long-range gravitational forces. Although the mean field dominates the dynamics, **weak, distant encounters** accumulate over time and produce gradual diffusion of energy and angular momentum.

Deflection by a single encounter: For a relative velocity v and impact parameter b, the deflection angle θ is

$$\theta = \frac{2Gm}{bv^2}, \tag{136}$$

valid for small-angle (weak) encounters ($\theta \ll 1$). **(See Binney 2008 for a complete derivation)**.

Many-body effect: The cumulative change of velocity from many encounters is obtained by integrating over all possible impact parameters $b_{\min}$ to $b_{\max}$:

$$\langle (\Delta v)^2 \rangle \propto \int_{b_{\min}}^{b_{\max}} \frac{1}{b}\, db = \ln\left(\frac{b_{\max}}{b_{\min}}\right) \equiv \ln \Lambda. \tag{137}$$

where the **Coulomb logarithm** ($\ln \Lambda$) quantifies the range of impact parameters contributing to cumulative gravitational scattering.

Physical Interpretation of the Coulomb Logarithm

Limits of integration:

- $b_{\min}$ — the smallest impact parameter, where the deflection becomes large ($\theta \sim 1$):

$$b_{\min} \simeq \frac{Gm}{v^2}. \tag{138}$$

- $b_{\max}$ — the largest distance over which encounters remain independent, typically of order the system size R or local scale height:

$$b_{\max} \simeq \eta R. \tag{139}$$

Typical value: For star clusters or galaxies, $\ln \Lambda \simeq 10$–20.

Definition and Physical Meaning of Relaxation Time

The **relaxation time** quantifies how long it takes a stellar system to experience significant energy redistribution due to cumulative weak gravitational encounters. **(See Binney 2008 for a complete derivation)**

Definition:

$$t_{\rm relax} \simeq \frac{0.1\,N}{\ln\Lambda}\,t_{\rm cross}, \tag{140}$$

where

- N is the number of particles (stars),
- Λ is the **Coulomb parameter**, $\Lambda \simeq b_{\rm max}/b_{\rm min} \simeq N$,
- $t_{\rm cross} \simeq R/\sigma \simeq R/v_c$ is the dynamical (crossing) time.

Physical meaning:

- Stars experience many small-angle deflections per crossing.
- Over $\sim N/(8\ln\Lambda)$ crossings, their velocity directions change substantially.
- Relaxation drives the system toward **stationary velocity distribution (e.g.Maxwellian)** and energy equipartition.

Relaxation Times for Different Stellar Systems

Order-of-magnitude examples:

System	N	R [pc]	σ [km/s]	t_{relax} [Gyr]
Open Cluster (e.g. Pleiades)	10^3	3	1	~ 0.1
Globular Cluster (e.g. 47 Tuc)	10^6	10	10	~ 1
Elliptical Galaxy (e.g. M87)	10^{11}	5000	250	$\sim 10^5$
Galaxy Cluster (e.g. Virgo)	10^{13}	10^6	1000	$\sim 10^8$

Scaling behavior:

$$t_{\mathrm{relax}} \propto \frac{N}{\ln N} \frac{R}{\sigma}. \tag{141}$$

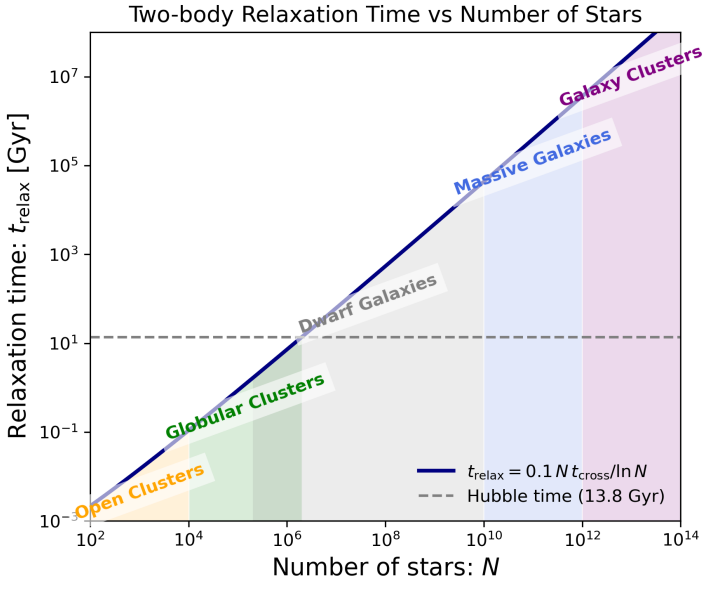
Interpretation:

- Small N and low $\sigma \rightarrow$ short relaxation (collisional regime).
- Large N and high $\sigma \rightarrow$ long relaxation (collisionless regime).

Collisional vs. Collisionless Regimes

1. **Star clusters (collisional systems):** $t_{\mathrm{relax}} \ll t_{\mathrm{Hubble}}$
   - Two-body encounters drive **core collapse** and **mass segregation**.
   - Energy exchange leads to **evaporation** and tidal loss.
2. **Galaxies (collisionless systems):** $t_{\mathrm{relax}} \gg t_{\mathrm{Hubble}}$
   - Dynamics governed by smooth mean field $\rightarrow$ **collisionless Boltzmann equation**.
   - Evolution by mergers **violent relaxation**, instabilities, and collective effects (bars, waves).
3. **Galaxy clusters:** $t_{\mathrm{relax}} \gg t_{\mathrm{Hubble}}$
   - Encounters negligible; Relaxation time exceeds age of Universe.
   - But! Relaxation via large-scale **violent relaxation** and **dynamical friction**.

Summary: $t_{\mathrm{relax}}(\mathrm{open\ cluster}) \ll t_{\mathrm{relax}}(\mathrm{globular}) \ll t_{\mathrm{relax}}(\mathrm{galaxy}) \ll t_{\mathrm{relax}}(\mathrm{cluster})$


Two-body Relaxation Time vs Number of Stars
Relaxation time: $t_{\rm relax}$ [Gyr]
Number of stars: N
10^{-3}
10^{-1}
10^{1}
10^{3}
10^{5}
10^{7}
10^{2}
10^{4}
10^{6}
10^{8}
10^{10}
10^{12}
10^{14}
Open Clusters
Globular Clusters
Dwarf Galaxies
Massive Galaxies
Galaxy Clusters
$t_{\rm relax} = 0.1\,N\,t_{\rm cross}/\ln N$
Hubble time (13.8 Gyr)

Is Dark Matter Ever Two-body Relaxed?

Setup (Milky Way–like halo):

- Total dark matter mass $M_{\rm DM} \simeq 10^{12}\, M_\odot$
- Crossing time $t_{\rm cross} \sim \dfrac{R}{V} \sim 0.2\ {\rm Gyr}$ (order-of-magnitude)
- Two-body relaxation:

$$t_{\rm relax} \;\simeq\; \frac{0.1\,N}{\ln N}\, t_{\rm cross} \tag{142}$$

Candidate particles and implied N and $t_{\rm relax}$:

Candidate	**Particle mass**	**Number** $N = \dfrac{M_{\rm DM}}{m_{\rm p}}$	$t_{\rm relax}$ [Gyr]
MACHO / PBH	$1\,M_\odot$	$\sim 10^{12}$	$\sim 7 \times 10^{8}$
WIMP	$100\ {\rm GeV}/c^2$	$\sim 1.1 \times 10^{67}$	$\sim 1 \times 10^{63}$
Axion	$10^{-5}\ {\rm eV}/c^2$	$\sim 1.1 \times 10^{83}$	$\sim 10^{79}$

Notes: $1\ {\rm GeV}/c^2 = 1.7827 \times 10^{-27}\,{\rm kg}$, $M_\odot = 1.9885 \times 10^{30}\,{\rm kg}$. Numbers use $\ln N$ in Eq. (142) and $t_{\rm cross} = 0.2$ Gyr.

Takeaway: Even in the most "collisional" toy case (all DM as $\sim 1\,M_\odot$ black holes), $t_{\rm relax} \gg$ Hubble time (13.8 Gyr). For particle DM (WIMPs, axions) the implied N makes $t_{\rm relax}$ *astronomically* larger.

Caveats & clarity:

- Dark matter is **(nearly) gravitationally collision-less** : phase-space evolution follows the CBE.
- Particle *annihilation* (e.g., WIMPs) can produce γ rays, but this is a **particle-physics** process, not two-body *gravitational* relaxation.
- MACHO/PBH[1]-dominated DM is strongly constrained; the table is a **thought experiment**.

[1]PBH:primordial black hole

Violent Relaxation: Rapid Collisionless Evolution

Question: How can galaxies reach quasi-equilibrium if $t_{\rm relax} \gg t_{\rm Hubble}$?

During collapse or mergers, the gravitational potential changes rapidly:

$$\boxed{\frac{d}{dt}\left(\frac{v^2}{2} + \Phi\right) = \frac{\partial \Phi}{\partial t}} \tag{143}$$

- A time-dependent potential redistributes **stellar orbital energies**.
- Orbital mixing produces a quasi-stationary system in a few dynamical times.
- **No two-body encounters are required**.

Examples: galaxy mergers, dark matter halo collapse, and early star-cluster assembly.

Lynden-Bell (1967), MNRAS 136, 101

Phase Mixing and Violent Relaxation

Collisionless evolution: The distribution function is conserved along stellar trajectories:

$$\boxed{\frac{Df}{Dt} = 0} \qquad \text{(Collisionless Boltzmann equation)} \tag{144}$$

Phase mixing: Stars on different orbits have different frequencies:

$$\Delta\theta(t) = \Delta\theta_0 + \Delta\Omega\, t \tag{145}$$

- Initially nearby stars spread in orbital phase.
- The distribution stretches and folds in phase space.
- Occurs even in a static potential ($E = \text{const}$).

Violent relaxation: During collapse or mergers, the potential changes rapidly.

- Individual stellar orbital energies change.
- Phase mixing accompanies the redistribution.
- A quasi-equilibrium can emerge in a few crossing times.

Both are collisionless processes: $Df/Dt = 0$ remains valid.

Internal Dynamical Processes in Star Clusters

Stellar Collisions

- Energy equipartition causes **Mass segregation**: massive stars/objects to sink to the center.
- Collisions affect binary formation/destruction, BH retention.

Core collapse

- Heat flows outward (gravothermal instability).
- Core shrinks, density rises; halted by binaries or BH subsystem.

Internal instabilities

- Radial-orbit instability in anisotropic systems.
- Collective modes (oscillations, gravothermal breathing).

Binary Stars as Energy Reservoirs

Internal energy of a bound binary:

$$E_{\rm int} = \frac{1}{2}\mu v_{\rm rel}^2 - \frac{Gm_1m_2}{r} \tag{146}$$

$$E_{\rm int} = -\frac{Gm_1m_2}{2a} \equiv -E_{\rm b}, \qquad E_{\rm b} > 0 \tag{147}$$

$\mu = m_1m_2/(m_1+m_2)$; a is the semi-major axis.

A binary–single encounter exchanges energy:

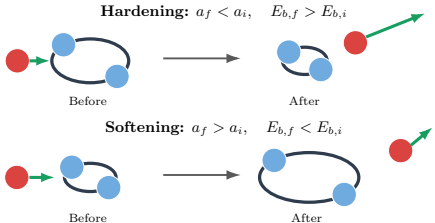
$$\Delta K_{\rm encounter} = -\Delta E_{\rm int} = \Delta E_{\rm b} \tag{148}$$

Heggie's law (statistical):

- **Hard binaries:** $E_{\rm b} \gg K_{\rm enc}$; typically $E_{\rm b}\uparrow$, $a\downarrow$; stars gain kinetic energy.
- **Soft binaries:** $E_{\rm b} \ll K_{\rm enc}$; typically $E_{\rm b}\downarrow$, $a\uparrow$; possible disruption.

$K_{\rm enc}$: incoming encounter kinetic energy.

Heggie (1975), MNRAS 173, 729



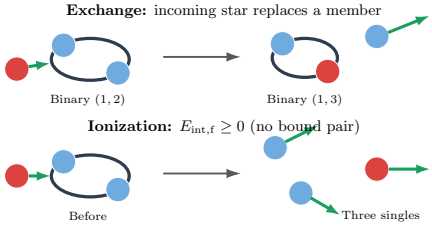
Schematic binary–single flybys.

Heggies's Law: Hard become harder, soft become softer.

Binary–Single Encounters: Exchange and Ionization

Main $1+2$ scattering outcomes:

- **Flyby:** the binary survives, but its orbital energy and eccentricity may change.
- **Exchange:** the incoming star replaces one binary member; retention of more massive stars is favored.
- **Ionization:** the binary is destroyed, leaving three unbound stars.



Schematic final states; individual outcomes are stochastic.

Energetic condition for ionization:

$$v_\infty \geq v_{\rm c}, \qquad v_{\rm c}^2 = \frac{G m_1 m_2 (m_1 + m_2 + m_3)}{a\, m_3 (m_1 + m_2)} \qquad (149)$$

v_∞: relative speed between star 3 and the binary centre of mass at infinity.

Note: $v_\infty \geq v_{\rm c}$ is necessary, *not sufficient*, for ionization. Binary–binary $(2+2)$ encounters also occur; physical collisions are distinct from gravitational scattering.

Hut & Bahcall (1983), ApJ 268, 319 | Hills & Fullerton (1980), AJ 85, 1281

Typical Velocities of Hard and Soft Binaries

For a circular binary with masses m_1, m_2 and semi-major axis a:

$$v_{\rm rel} = \sqrt{\frac{G(m_1+m_2)}{a}}, \qquad v_\star = \frac{v_{\rm rel}}{2} \quad (m_1 = m_2). \tag{150}$$

Example: $m_1 = m_2 = 1\,M_\odot$, in a globular cluster with $\sigma = 10$ km s^{-1}.

Separation a	$v_\star$ (km s^{-1})	Type
0.1 AU	66.6	Very hard
1 AU	21.1	Hard
10 AU	6.7	Near boundary
100 AU	2.1	Soft
1000 AU	0.7	Very soft

Hard–soft boundary: Binding energy comparable to the kinetic energy of typical cluster stars.

- Globular cluster ($\sigma \sim 10$ km/s): $a_{\rm hs} \sim 9$ AU.
- Open cluster ($\sigma \sim 1$ km/s): $a_{\rm hs} \sim 900$ AU.

Unresolved Binaries: Velocity Dispersion Bias

Observed radial velocity:

$$v_{\rm obs} = v_{\rm COM} + v_{\rm orb} + \epsilon \qquad (151)$$

Unresolved orbital motions inflate the observed dispersion:

$$\sigma^2_{\rm obs} \simeq \sigma^2_{\rm cluster} + f_b \langle v^2_{\rm orb,los} \rangle \qquad (152)$$

How to correct it:

- Multi-epoch radial velocities.
- Fit binary orbits ($v_{\rm COM}$).
- Model undetected binaries statistically.

Numerical example:
$\sigma_{\rm cluster} = 2$ km/s, $f_b = 0.5$,
$\langle v^2_{\rm orb,los} \rangle^{1/2} = 3$ km/s.

$$\sigma_{\rm obs} = \sqrt{2^2 + 0.5(3^2)} = 2.92 \ {\rm km/s} \qquad (153)$$

Since $M_{\rm dyn} \propto \sigma^2$:

$$\boxed{\frac{M_{\rm dyn,obs}}{M_{\rm true}} \simeq \frac{2.92^2}{2^2} \simeq 2.1} \qquad (154)$$

Result: Dynamical mass overestimated by a factor ~ 2.

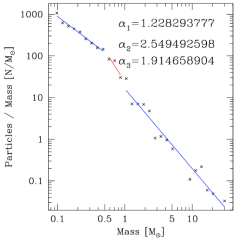
Applications: Young star clusters (large biases); dwarf spheroidals (typically smaller biases).

Gieles et al. (2010) | Minor et al. (2010)

IMF, Mass Spectrum, and Equipartition

Initial Mass Function (IMF):

- Clusters are born with a broad mass distribution (Kroupa/Chabrier-like).
- Typical range: 0.1–50 $M_\odot$.
- Massive stars are rare but dominate energetic encounters.



Energy exchange $\Rightarrow$ equipartition:

$$K_1 = K_2 \quad \Rightarrow \quad \frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_2 v_2^2$$

Given $m_2 = 4m_1$:

$$v_2 = \frac{v_1}{2}, \qquad v \propto m^{-1/2}$$

Consequences:

- Massive stars lose kinetic energy $\Rightarrow$ sink to center.
- Low-mass stars gain kinetic energy $\Rightarrow$ move outward.
- Classical mechanism behind **mass segregation**.

Spitzer Instability: When Equipartition Fails

Consider light stars (m_1, M_1) and a population of heavier stars (m_2, M_2).

Spitzer's criterion for full equipartition:

$$\boxed{\frac{M_2}{M_1}\left(\frac{m_2}{m_1}\right)^{3/2} \lesssim 0.16} \tag{155}$$

Example: if $m_2 = 10m_1$, equipartition requires $M_2/M_1 \lesssim 0.005$.

If the heavy population is too abundant:

- Full equipartition cannot be maintained.
- Massive stars or remnants concentrate in a self-gravitating central subsystem.
- The subsystem can undergo rapid contraction, enhancing binary formation and strong encounters.

Consequence: Mass segregation can proceed without complete thermalization.

Spitzer (1969); Trenti & van der Marel (2013)

Mass Segregation Simulation

(Download the PDF and open with Adobe Acrobat Reader to play video.)

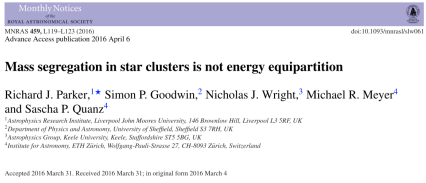
Mass Segregation Without Equipartition? (Parker et al. 2016)

Classical view:

- Mass segregation caused by two-body relaxation.
- Energy equipartition: $mv^2 \approx$ const.
- Massive stars drift inward as they lose kinetic energy.

Parker et al. (2016):

- Young clusters can appear mass segregated *before* relaxation can operate.
- Initial fractal/substructured distributions can collapse and **naturally concentrate massive stars**.
- Mass segregation may be **primordial**, not equipartition-driven.



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Mass segregation in star clusters is not energy equipartition

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ABSTRACT

Mass segregation in star clusters is often thought to indicate the onset of energy equipartition, where the most massive stars impart kinetic energy to the lower-mass stars and brown dwarfs/free-floating planets. The predicted net result of this is that the centrally concentrated massive stars should have significantly lower velocities than fast-moving low-mass objects on the periphery of the cluster. We search for energy equipartition in initially spatially and kinematically substructured N-body simulations of star clusters with $N = 1500$ stars, evolved for 100 Myr. In clusters that show significant mass segregation we find no differences in the proper motions or radial velocities as a function of mass. The kinetic energies of all stars decrease as the clusters relax, but the kinetic energies of the most massive stars do not decrease faster than those of lower-mass stars. These results suggest that dynamical mass segregation – which is observed in many star clusters – is not a signature of energy equipartition from two-body relaxation.

Key words: methods: numerical – stars: formation – open clusters and associations: general.

Parker et al. 2016, MNRAS 459, L119

http://arxiv.org/abs/1604.00394

Gravothermal Instability: Negative Heat Capacity

A self-gravitating system can exhibit **negative heat capacity** (contrary to a hot gas where $T \propto K$).
From the virial theorem, for an approximately self-gravitating virialized subsystem:

$$2K + W \simeq 0, \qquad E = K + W \simeq -K. \tag{156}$$

Defining an effective kinetic temperature $K = \frac{3}{2} N k_{\rm B} T$:

$$E \simeq -\frac{3}{2} N k_{\rm B} T, \qquad \boxed{C = \frac{dE}{dT} < 0}. \tag{157}$$

Physical consequence:

- The core transfers energy to the surrounding halo.
- Losing energy causes the core to **contract and become hotter**.
- The increased temperature contrast drives further outward energy transport.

The simple virial argument neglects surface-pressure and interaction terms of the embedded core.

Lynden-Bell & Wood (1968)

Gravothermal Core Collapse

The self-amplifying process:

1. Two-body relaxation transports energy from the dense core to the halo.
2. The core loses energy, contracts, and increases its velocity dispersion.
3. Its density rises and central relaxation time becomes shorter.
4. Energy transport accelerates, driving further contraction (runaway processes).

What eventually halts core collapse?

- Hard binaries release binding energy through three-body encounters.
- Binary heating can drive post-collapse expansion.
- Stellar mass loss can also modify the core's evolution.

Important: Core collapse does not require physical stellar collisions and can occur even in equal-mass clusters.

Stellar Encounters & Cluster Evaporation

Star Cluster Evaporation

Two-body relaxation causes gradual diffusion in orbital energy. Some stars acquire sufficient energy to escape.

For an isolated cluster with $\Phi(\infty) = 0$:

$$\varepsilon = \frac{1}{2}v^2 + \Phi(r) \tag{158}$$

$$\boxed{v_{\rm esc}(r) = \sqrt{-2\Phi(r)}} \tag{159}$$

- Stars escape when $\varepsilon > 0$.
- Relaxation replenishes the high-energy tail of the stellar velocity distribution.
- Escaping stars gradually reduce cluster mass.

For a virialized isolated cluster, the mass-weighted average is

$$\langle v_{\rm esc}^2 \rangle_M = 4\langle v^2 \rangle_M. \tag{160}$$

This global relation does not imply a spatially constant escape velocity.

Evaporation: Cluster Mass Loss

A simple relaxation-driven mass-loss prescription:

$$\boxed{\frac{dM}{dt} \simeq -\xi_e \frac{M}{t_{\rm rh}}} \tag{161}$$

where ξ_e is the fraction of cluster mass lost per half-mass relaxation time.

- **Isolated clusters:** evaporation is very slow ($t_{\rm evap} \gg t_{\rm rh}$).
- **Tidally limited clusters:** the Galactic tidal field lowers the energy needed for escape.
- **Potential escapers:** some energetically unbound stars remain near the cluster for many crossing times.

Distinction: Evaporation is driven by internal relaxation; tidal stripping and shocks are external processes.

The evaporation rate depends on cluster structure, stellar masses, and the tidal environment.

Environmental Dynamical Processes Acting on Clusters

Clusters also evolve due to **external forces** from their host galaxy:

Tidal stripping

- Stars beyond the tidal radius become unbound.
- Strong in pericenter passages; produces **tidal tails/streams**.

Tidal heating / compression / shocking

- Disk crossings, bulge passages, GMC encounters inject energy.
- Can puff up or compress the cluster depending on geometry.

External tidal fields

- Shape cluster structure and tidal boundary.
- Impose non-spherical equipotential surfaces (Roche lobes).

Tidal Stripping and the Jacobi (Hill) Radius

Tidal stripping occurs when a satellite (cluster or dwarf galaxy) orbits inside the potential of a more massive host. Stars beyond a critical radius become unbound due to the host's tidal field.

Jacobi (Hill) radius for a satellite of mass m orbiting a host of mass $M(< R)$:

$$r_J = R \left(\frac{m}{3\, M(< R)} \right)^{1/3}, \tag{162}$$

derived in the rotating frame by balancing:

$$\text{gravity of satellite} \quad = \quad \text{tidal} + \text{centrifugal forces}.$$

This radius is also called the **tidal radius** or **Hill radius**.

King-model "tidal radius" is different:

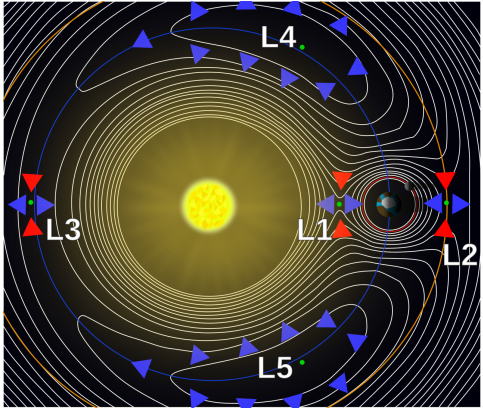
- The King *tidal radius* ($r_t^{\rm King}$) is a free parameter from surface-brightness profile fitting.
- It does *not* necessarily correspond to the dynamical Jacobi radius, which can be larger or smaller than the true dynamical truncation radius depending on the orbital position.

Lagrange Points and the Tidal Boundary

In the rotating frame of a satellite orbiting a host galaxy, the effective potential has five equilibrium points, the **Lagrange points:**

- Inner (L_1) and outer (L_2) saddle points (escape channels). Tidal arms are generated from here, that can be leading (L_1) or trailing (L_2).
- L_3: Opposite side of host; minor role.
- L_4: Leading triangular point (quasi-stable, Trojans(Greeks)).
- L_5: Trailing triangular point (quasi-stable, Trojans).

Key idea: L_1 and L_2 define the **Jacobi radius** and dominate tidal stripping.



Source:Wiki

Modern perspective: Tidal truncation depends on the **full tidal tensor**, orbital phase, mass distribution, and pericentric passages.
Reference: Blaña et al. (2025), *Tidal truncation of satellite galaxies in realistic cluster potentials*.
https://ui.adsabs.harvard.edu/#abs/2025A%26A...699A..69B/abstract We show that the effective tidal radius differs from simple Jacobi estimates and evolves strongly along the orbit.
It is possible to re-write the Jacobi radius as a density relation:

$$\bar{\rho}_{\rm sat} \geq 3\bar{\rho}_{\rm host} \tag{163}$$

Where the host's tidal field ($\tau = 3\bar{\rho}_{\rm host}$) is invertionally proportional to the distance

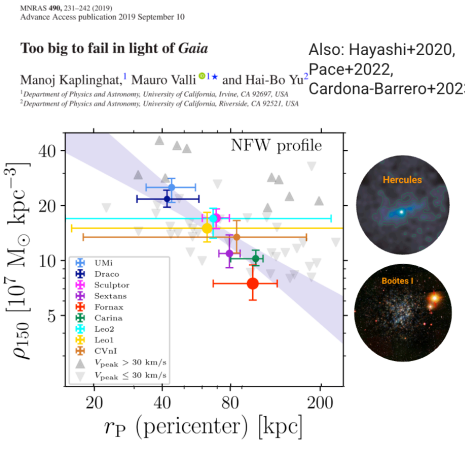
$$\bar{\rho}_{\rm host} \propto (1/r) \tag{164}$$

And so,

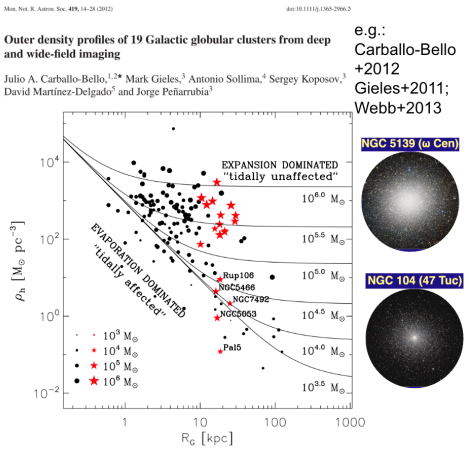
$$\bar{\rho}_{\rm sat} \propto 1/R_{\rm peri} \tag{165}$$

Mass Density vs Rperi

Relation for dwarfs and star clusters (example. Carballo-Bello+2012: https://academic.oup.com/mnras/article/419/1/14/998277)



Also: Hayashi+2020, Pace+2022, Cardona-Barrero+2023,



e.g.: Carballo-Bello +2012 Gieles+2011; Webb+2013

Tidal Stripping Simulation

(Download the PDF and open with Adobe Acrobat Reader to play video.)

Tidal Shocking of Star Clusters

When a star cluster passes rapidly through a region where the gravitational field changes steeply (e.g., pericentre, disk crossing, bar passage), a star receives an impulsive energy injection or "tidal shock".

Energy gain per unit mass in the impulsive approximation (from **Hammer et al. 2024, MNRAS, 527, 2718**):

$$\Delta\epsilon = \frac{4}{3}\,\eta \left(\frac{G\,M_{\rm Host}(R_{\rm peri})}{R_{\rm peri}^2} \right)^2 \left(\frac{r_{\rm rms}}{V_{\rm peri}} \right)^2, \tag{166}$$

where: $V_{\rm peri}$ = cluster orbital velocity at pericentre, $r_{\rm rms}$ = 3D rms radius of the stellar system, η accounts for deviations from the ideal impulsive limit.

Physical consequences:

- Cluster heats and expands after each shock.
- Repeated shocking accelerates mass loss and can trigger disruption.

Extreme Tidal Shocking (Disruption Limit)

In the most extreme scenario, when the impulsive tidal energy input from the host (Milky Way) is larger or equal to the binding energy per unit mass of a stellar system ($|u| = K/M = \frac{3}{2}\sigma^2$), the tidal shocking reaches the **disruption limit ($\Delta\epsilon/|u| \geq 1$)**. Combining equations for $\Delta\epsilon$, and $r_{\rm rms}^2 = 3\, r_{\rm half,3D}^2$, and $r_{\rm half,3D} = 4/3\, r_{\rm half}$ yields the disruption-condition:

$$\frac{\sigma_{\rm l}^2}{r_{\rm half}^2} = \frac{128}{27}\,\eta\left(\frac{g_{\rm host}(R_{\rm peri})}{V_{\rm peri}}\right)^2, \tag{167}$$

where: $\sigma_{\rm l}$ = one-dimensional (un-binding limit) velocity dispersion of the satellite (dwarf or star cluster), $V_{\rm peri}$ = orbital velocity at pericentre, the gravitational field $g_{\rm host}(R_{\rm peri}) = GM_{\rm host}(R_{\rm peri})\, R_{\rm peri}^{-2}$, and η encodes deviations from a pure impulsive shock.

Interpretation: clusters with low $|u|$ (or low initial σ) and a large $r_{\rm half}$ are highly vulnerable to tidal disruption during pericentre passages.

Tidal Stripping Simulation: tidal shock

(Download the PDF and open with Adobe Acrobat Reader to play video.)

Environmental Dynamical Processes Acting on Clusters

Dynamical friction

- Clusters lose orbital energy to the background stellar/DM field.
- Leads to orbital decay into the galactic center (Bulge NSCs).

Ram pressure (gas-rich environments)

- Relevant in gas-rich hosts or during early formation.
- Removes gas; affects embedded cluster survival.

Case Study I: Baumgardt & Makino (2003)

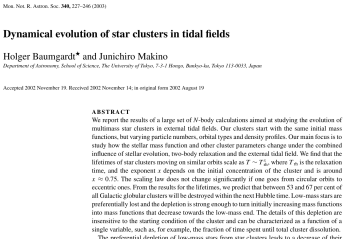
Question: How do relaxation and Galactic tides set cluster lifetimes?

Method:

- Direct collisional `NBODY4` simulations on GRAPE-6 ($N = 8192$–131072 stars).
- Multimass King models ($W_0 = 5, 7$), Kroupa IMF, stellar evolution.
- Circular/eccentric orbits in a fixed Galactic logarithmic potential.

Combined physics: two-body relaxation, mass segregation, stellar mass loss and tidal escape.

NASA ADS | Full paper (arXiv PDF)



Mon. Not. R. Astron. Soc. **340**, 227–246 (2003)

Dynamical evolution of star clusters in tidal fields

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ABSTRACT

We report the results of a large set of N-body calculations aimed at studying the evolution of multimass star clusters in external tidal fields. Our clusters start with the same initial mass functions, but varying particle numbers, orbital types and density profiles. Our main focus is to study how the stellar mass function and other cluster parameters change under the combined influence of stellar evolution, two-body relaxation and the external tidal field. We find that the lifetimes of star clusters moving on similar orbits scale as $T \sim T_{\rm rh}^{x}$, where $T_{\rm rh}$ is the relaxation time, and the exponent x depends on the initial concentration of the cluster and is around $x \approx 0.75$. The scaling law does not change significantly if one goes from circular orbits to eccentric ones. From the results for the lifetimes, we predict that between 53 and 67 per cent of all Galactic globular clusters will be destroyed within the next Hubble time. Low-mass stars are preferentially lost and the depletion is strong enough to turn initially increasing mass functions into mass functions that decrease towards the low-mass end. The details of this depletion are insensitive to the starting condition of the cluster and can be characterized as a function of a single variable, such as, for example, the fraction of time spent until total cluster dissolution.

The preferential depletion of low-mass stars from star clusters leads to a decrease of their

First-page excerpt, MNRAS 340, 227.

Baumgardt & Makino (2003): Cluster Dissolution

Figure 3: dissolution time versus initial cluster mass.

- More massive clusters generally survive longer.
- Stronger Galactic tides shorten their lifetimes.
- At a fixed orbit, the dependence on relaxation time is sub-linear.

$$T_{\rm diss} = k\, t_{\rm rh}^{x}\, t_{\rm cross}^{1-x} \qquad (168)$$

$x \simeq 0.75$ ($W_0 = 5$); 0.82 ($W_0 = 7$).

Why sub-linear? Potential escapers can remain temporarily trapped before crossing the tidal boundary.

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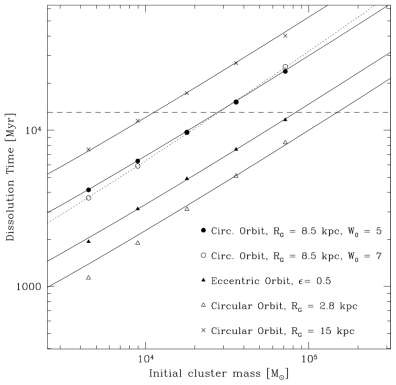


Fig. 3: $T_{\rm diss}$ versus $M_{\rm init}$; dashed line: 13 Gyr.

Case Study II: Kruijssen et al. (2011)

Question: How does an evolving galaxy shape its entire star-cluster population?

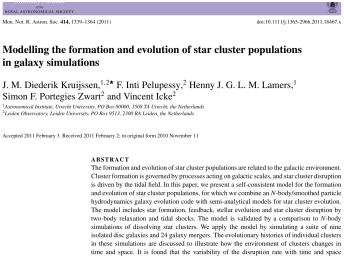
Method:

- Galaxy evolution with *N*-body + SPH (stars, gas, dark matter, feedback).
- Star clusters treated **semi-analytically as subgrid objects**, not as resolved stellar *N*-body systems.
- Nine isolated disc galaxies and 24 mergers.

Mass-loss channels:

$$\dot{M} = \dot{M}_{\rm ev} + \dot{M}_{\rm rlx} + \dot{M}_{\rm sh} \qquad (169)$$

NASA ADS | Full paper (arXiv PDF)



MONTHLY NOTICES OF THE ROYAL ASTRONOMICAL SOCIETY

Mon. Not. R. Astron. Soc. **414**, 1339–1364 (2011) doi:10.1111/j.1365-2966.2011.18467.x

Modelling the formation and evolution of star cluster populations in galaxy simulations

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ABSTRACT

The formation and evolution of star cluster populations are related to the galactic environment. Cluster formation is governed by processes acting on galactic scales, and star cluster disruption is driven by the tidal field. In this paper, we present a self-consistent model for the formation and evolution of star cluster populations, for which we combine an *N*-body/smoothed particle hydrodynamics galaxy evolution code with semi-analytical models for star cluster evolution. The model includes star formation, feedback, stellar evolution and star cluster disruption by two-body relaxation and tidal shocks. The model is validated by a comparison to *N*-body simulations of dissolving star clusters. We apply the model by simulating a suite of nine isolated disc galaxies and 24 galaxy mergers. The evolutionary histories of individual clusters in these simulations are discussed to illustrate how the environment of clusters changes in time and space. It is found that the variability of the disruption rate with time and space

First-page excerpt, MNRAS 414, 1339.

Kruijssen et al. (2011): Tides and Cluster Survival

Figure 6: two clusters with different orbits within an evolving disc.

- **Top:** cluster mass $M(t)$.
- **Middle:** tidal-field strength.
- **Bottom:** integrated shock heating.

Tidal tensor:

$$T_{ij} = -\frac{\partial^2 \Phi}{\partial x_i \partial x_j} \qquad (170)$$

Result: the tidal environment varies along orbits; rapid mass-loss episodes correlate with shocks.

In their disc models, tidal shocks account for roughly 80–85% of disruption.

NASA ADS | Full paper (arXiv PDF)

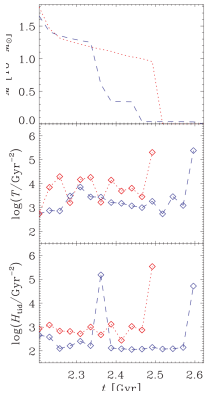
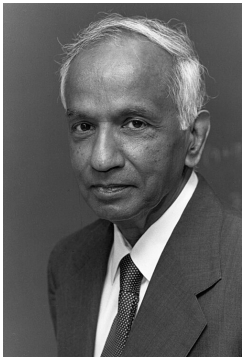


Fig. 6: mass, tidal-field strength and shock heating.

Chandrasekhar's

Dynamical Friction



Subrahmanyan Chandrasekhar (Wiki)

Chandrasekhar's Dynamical Friction

Context: A massive object of mass M moves through a sea of lighter background particles of density ρ and isotropic velocity distribution $f(v)$. Gravitational encounters with these particles deflect them and produce a density enhancement behind M, which exerts a drag force opposing its motion (a gravitational *wake*).

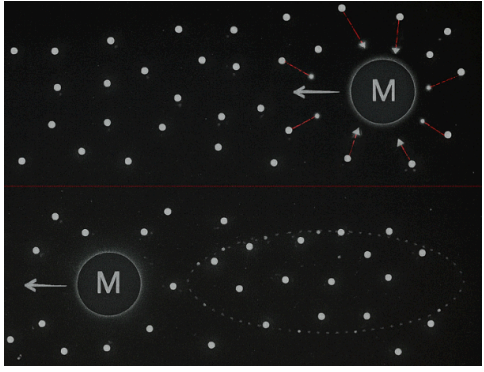
$$\frac{d\vec{v}_M}{dt} = -4\pi\, G^2\, M\rho\, \ln\Lambda\, \frac{\mathbf{v}_M}{v_M^3} \left[\mathrm{erf}(X) - \frac{2X}{\sqrt{\pi}}\, e^{-X^2}\right] \quad \text{with} X \equiv \frac{v_M}{\sqrt{2}\,\sigma} \tag{171}$$

Parameters: M — mass of the perturber, ρ — local density of background particles, σ — 1D velocity dispersion of background, $\ln\Lambda$ — Coulomb logarithm, $\Lambda = b_{\max}/b_{\min}$.

Physical Picture of the Gravitational Wake

The gravitational wake:

- The moving mass M deflects background particles, leaving an overdensity trailing behind it.
- The combined gravitational pull of this wake acts as a **drag force**.
- The direction of the friction is opposite to $\vec{v}_M$.



Interpretation and Scaling of Dynamical Friction

- The drag force $\propto M^2$: more massive objects decelerate faster.
- For $v_M \gg \sigma$: only slow particles contribute $\Rightarrow$ friction $\propto 1/v_M^2$.
- For $v_M \lesssim \sigma$: nearly all particles contribute $\Rightarrow$ friction saturates $\propto v_M$.

Energy loss rate:

$$\frac{dE}{dt} = M\mathbf{v}_M \cdot \frac{d\mathbf{v}_M}{dt} \simeq -4\pi\, G^2\, M^2\, \rho\, \ln\Lambda\, \frac{1}{v_M} \left[\mathrm{erf}(X) - \frac{2X}{\sqrt{\pi}}\, e^{-X^2}\right] \tag{172}$$

Characteristic decay timescale:

$$t_{\rm df} \sim K/(dE/dt) \sim \frac{v_M^3}{4\pi\, G^2\, M\rho\, \ln\Lambda} \tag{173}$$

Interpretation: for a satellite galaxy or massive black hole orbiting in a galactic halo, $t_{\rm df}$ gives the time to spiral inward and merge with the center (typically $\sim$ few Gyr for $M \gtrsim 10^9\, M_\odot$).

Orbital dynamics

Orbital Dynamics: Overview

Orbital dynamics studies the motion of test particles or stars in a prescribed gravitational potential $\Phi(\mathbf{x})$.

Key ingredients:

- The potential $\Phi(\mathbf{x}, t)$ (static or rotating).
- Integration of equations of motion:

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\nabla\Phi.$$

- Conserved quantities (energy, angular momentum) depending on symmetry.
- Classification of orbits (tubes, boxes, resonant families).

Applications: galaxy disks, bars, halos, globular clusters, streams, tidal disruption, dark-matter halos.

Analytical Potentials Commonly Used

Plummer Sphere:

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}}. \tag{174}$$

NFW Halo:

$$\Phi_{\mathrm{NFW}}(r) = -4\pi G\rho_0 r_s^3 \frac{\ln(1 + r/r_s)}{r} \tag{175}$$

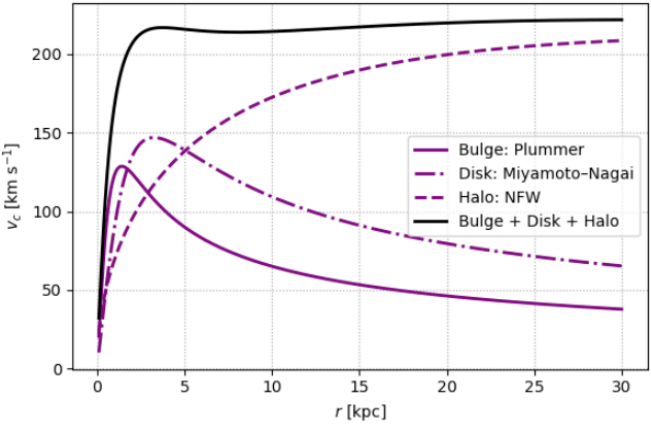
Miyamoto–Nagai Disk:

$$\Phi(R, z) = -\frac{GM}{\sqrt{R^2 + \left(a + \sqrt{z^2 + b^2}\right)^2}}. \tag{176}$$

Use cases:

- Plummer $\rightarrow$ star clusters, or galaxy bulges
- MN $\rightarrow$ thin/thick stellar disks
- NFW $\rightarrow \Lambda$CDM halos

Figure: Rotation Curves (MN + NFW)



Computing Orbits: Leapfrog and Symplectic Methods

Equations of motion:

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\nabla\Phi.$$

Leapfrog (Kick–Drift–Kick): a symplectic integrator:

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{\Delta t}{2}\,\mathbf{a}(\mathbf{x}_n), \tag{177}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\,\mathbf{v}_{n+1/2}, \tag{178}$$

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{\Delta t}{2}\,\mathbf{a}(\mathbf{x}_{n+1}). \tag{179}$$

Why leapfrog?

- Symplectic $\rightarrow$ conserves energy better over long times (Hamiltonian canonical transformation: kinetic and potential operators).
- Ideal for galaxy and cluster dynamics.

Choosing Δt: at least 1/10 of the crossing/orbital time (depends on velocity and acceleration as well).

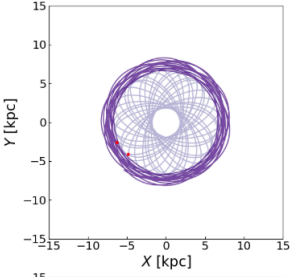
Orbitrs and Conserved Quantities in Different Potentials

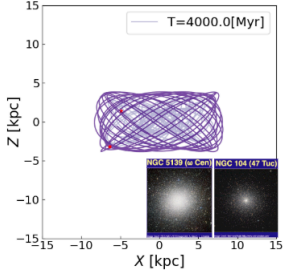
Different potentials support different orbit types:
Spherical Potentials:

- Energy E and Angular momentum vector **L** conserved.
- Motion confined to a plane.
- Circular orbits, Rosette orbits (not closed)
- Closed orbits only for special potentials (Kepler, Isochrone, Harmonic).

Axisymmetric disk potentials:

- Energy E conserved.
- Only L_z conserved.
- Circular and Tube orbits.





Barred Potentials and Rotating Frames

Galactic bars require a **rotating** non-axisymmetric potential:

$$\Phi(R, \phi, z, t) = \Phi(R, \phi - \Omega_p t, z).$$

Examples of bar potentials:

- Long & Murali (1992) bar.
- Dehnen bar potential.
- **Portail 2017, Sormani et al. (2022)** Milky Way bar (state-of-the-art).

Orbital families in bars:

- x_1 family (supports the bar).
- x_2 family (inner perpendicular orbits).
- Higher-order resonances (Banana, pretzel, etc.) build the box/peanut bulges
- **Chaotic orbits** appear in triaxial or barred galaxies.

In rotating frames (bars): the **Jacobi energy** E_J is conserved:

$$E_J = E - \Omega_p L_z \tag{180}$$

Codes for Orbit Integration

`DELOREAN` (Blaña et al. 2020+):

- Multi-component galaxy potentials.
- Leapfrog integrator, cosmic expansion motion equations.
- Python code, parallel orbit sampling, clusters + galaxies.

GALPY (Bovy 2015):

- Large library of analytic potentials.
- Orbit integration, actions, distribution functions.
- Python, widely used.

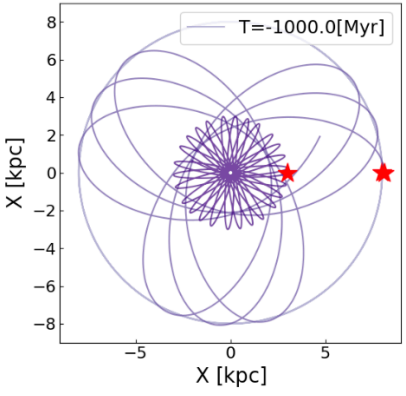
AGAMA (Vasiliev 2018):

- Fast action–angle methods.
- Self-consistent models, Schwarzschild.
- C++, Python bindings.

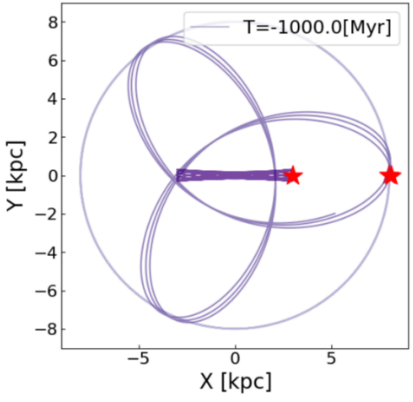
Example Orbits Computed with DELOREAN

Comparison of orbits with the same x,v IC., halo, disk, but spherical bulge vs bar (non-rotating).

Sphe. Bulge + Disk + Dark Halo (axi-symmetric pot.)



Bar + Disk + Dark Halo (non-axisymmetric pot.)



Example Orbit Computed with DELOREAN

Model parameters:

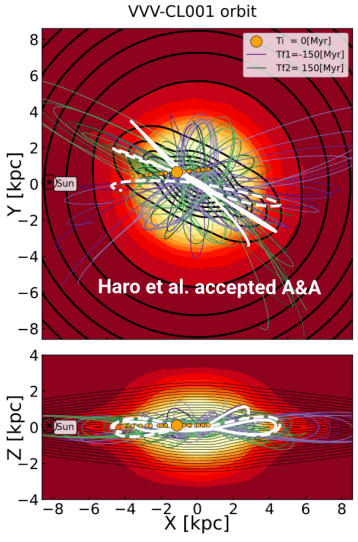
- Disk: Miyamoto–Nagai (M, a, b).
- Halo: NFW (ρ_s, r_s).
- Bulge/Bar: Box/peanut bulge + bar (Portail+2017, Sormani+2022).
- Pattern speed (bar): $\Omega_p = 40\,\mathrm{km/s/kpc}$.

Orbit properties:

- Energy E, An.Mom.L_z, Orb. Period T_R, T_ϕ, T_z.
- Apocentre r_a, pericentre:r_p, eccentricity: $\epsilon = (r_\mathrm{a} - r_\mathrm{p})(r_\mathrm{a} + r_\mathrm{p})^{-1}$

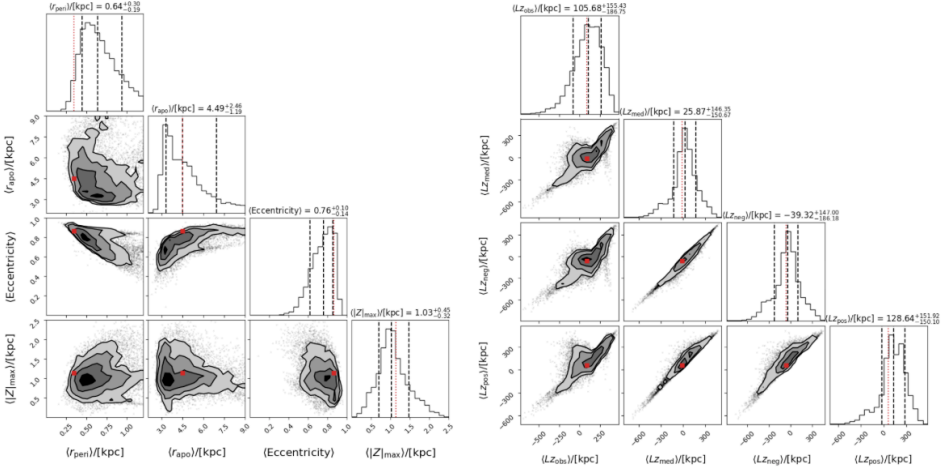
Reference: Blaña et al. 2020 MNRAS 497, DELOREAN.

Example of a star cluster orbit integrated in DELOREAN in a Milky Way potential with a bar, disk, gas, and dark matter halo.



Example Orbit Computed with DELOREAN: orbital properties

Statistics of 10^4 orbits (-5< t/Gyr <1).

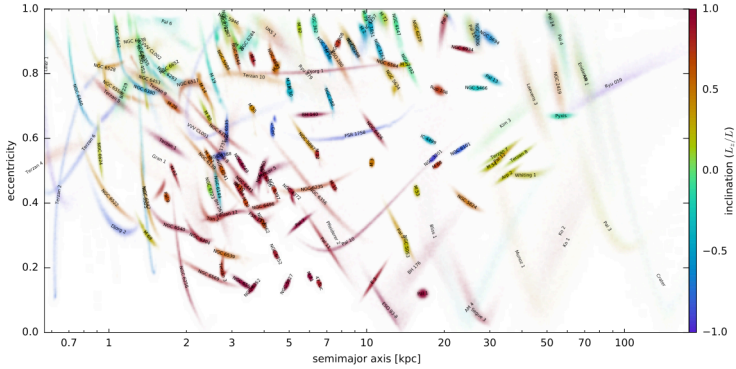


Star Clusters and stellar stream formation (DELOREAN-II Blaña in prep.)

(Download the PDF and open with Adobe Acrobat Reader to play video.)

Orbital properties

Example Vasiliev & Baumgardt 2021 MNRAS 505 4
https://ui.adsabs.harvard.edu/abs/2021MNRAS.505.5978V/abstract
Globular cluster orbital properties in an axisymmetric MW potential, but Lz is not conserved in barred potentials. (Note: here prograde = Lz>0 km/s kpc).



Exercise 1: Proper Motion $\rightarrow$ Tangential Velocity

Consider Omega Centauri with:

$$D_{\Omega\mathrm{Cen}} \simeq 5.2 \text{ kpc}, \qquad \mu \simeq 8 \text{ mas/yr}.$$

Tasks:

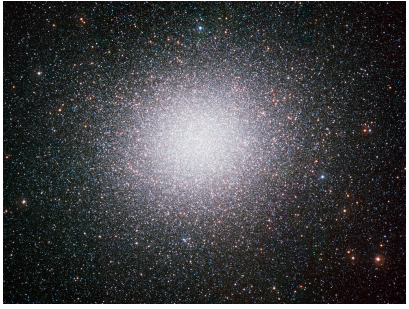
(a) Derive the relation between proper motion μ (mas/yr), distance D (kpc), and tangential velocity $V_{\tan}$ (km/s).

(b) Compute $V_{\tan}$ for:

$$D = 5.2,\ 52,\ 520 \text{ kpc}$$

assuming the *same* proper motion $\mu = 8$ mas/yr.

Goal: see how the same μ corresponds to very different $V_{\tan}$ at different distances.



Omega Centauri (Wiki, ESO)

Exercise 1: Solution and Distance Scaling

The standard conversion is

$$V_{\tan}\ [\mathrm{km/s}] = 4.74\,\mu\ [\mathrm{mas/yr}]\,D\ [\mathrm{kpc}]. \tag{181}$$

For $\mu = 8$ mas/yr:

$$D = 5.2\ \mathrm{kpc}: \qquad V_{\tan} \simeq 4.74 \times 8 \times 5.2 \approx 197\ \mathrm{km/s}, \tag{182}$$

$$D = 52\ \mathrm{kpc}: \qquad V_{\tan} \simeq 10 \times 197 \approx 2000\ \mathrm{km/s}, \tag{183}$$

$$D = 520\ \mathrm{kpc}: \qquad V_{\tan} \simeq 100 \times 197 \approx 2.0 \times 10^4\ \mathrm{km/s}. \tag{184}$$

Key point: a given proper motion corresponds to a tangential velocity that scales linearly with distance.
In practice: use `astropy.coordinates.SkyCoord` to convert between distance, proper motions, and 3D velocities.

Exercise 2: From $\sigma_{\rm los}$ to Proper-Motion Dispersion

For Omega Centauri, take approximate values:

$$D \simeq 5.2\ \mathrm{kpc}, \qquad \sigma_{\rm los} \simeq 13\ \mathrm{km\,s^{-1}}.$$

Assume:

- The velocity distribution is approximately isotropic, so the tangential dispersion is similar: $\sigma_t \approx \sigma_{\rm los}$.
- You observe a sample of member stars in proper motion.

Tasks:

(a) Write the relation between σ_t (km/s), D (kpc), and the proper-motion dispersion σ_μ (mas/yr).

(b) Estimate σ_μ for Omega Centauri.

Exercise 2: Solution (Omega Cen Sample)

Use the same conversion, applied to the *dispersion*:

$$\sigma_\mu \ [\mathrm{mas/yr}] = \frac{\sigma_t \ [\mathrm{km/s}]}{4.74 \, D \ [\mathrm{kpc}]}, \tag{185}$$

with $\sigma_t \approx \sigma_{\mathrm{los}} = 13 \ \mathrm{km/s}$ and $D = 5.2 \ \mathrm{kpc}$.

$$\sigma_\mu \approx \frac{13}{4.74 \times 5.2} \ \approx \ 0.53 \ \mathrm{mas/yr}. \tag{186}$$

Interpretation:

- The proper motions of member stars should have a spread of ~ 0.5 mas/yr due to internal motions.
- This can be directly compared with Gaia or HST proper-motion uncertainties when modeling the cluster's dynamics.

Again: `astropy.coordinates.SkyCoord` can perform these conversions with full 3D geometry and reference frames.

Exercise 3: Gaia DR3 vs Internal Motions of Omega Cen

From the previous exercise, Omega Centauri has an internal proper-motion dispersion of order:

$$\sigma_\mu^{(\mathrm{int})} \sim 0.5\ \mathrm{mas/yr}.$$

Assume that for bright cluster members in Gaia DR3 the typical 1D proper-motion uncertainty is:

$$\sigma_\mu^{(\mathrm{Gaia})} \sim 0.05\ \mathrm{mas/yr}.$$

Tasks:

(a) Compute the signal-to-noise ratio (SNR) for detecting internal motions in *one* star:

$$\mathrm{SNR}_\star = \frac{\sigma_\mu^{(\mathrm{int})}}{\sigma_\mu^{(\mathrm{Gaia})}}.$$

(b) If you average over N stars, the error on the *mean* motion scales as $\sigma_\mu^{(\mathrm{Gaia})}/\sqrt{N}$. For $N = 100$, what is the SNR on the mean?

(c) Qualitatively: is Gaia DR3 sufficient to resolve the internal kinematics of Omega Cen?

Exercise 3: Solution and Interpretation

(a) Single-star SNR:

$$\mathrm{SNR}_\star = \frac{\sigma_\mu^{(\mathrm{int})}}{\sigma_\mu^{(\mathrm{Gaia})}} \approx \frac{0.5}{0.05} \approx 10.$$

So each bright member has internal motions detected at $\sim 10\sigma$.

(b) SNR of the mean for $N = 100$ stars:

$$\sigma_{\mu,\mathrm{mean}}^{(\mathrm{Gaia})} = \frac{\sigma_\mu^{(\mathrm{Gaia})}}{\sqrt{N}} = \frac{0.05}{\sqrt{100}} = 0.005 \ \mathrm{mas/yr},$$

$$\mathrm{SNR}_{\mathrm{mean}} = \frac{\sigma_\mu^{(\mathrm{int})}}{\sigma_{\mu,\mathrm{mean}}^{(\mathrm{Gaia})}} \approx \frac{0.5}{0.005} \approx 100.$$

(c) Interpretation:

- Gaia DR3 can resolve internal kinematics of Omega Cen with very high significance.
- Combining many stars allows detailed mapping of velocity-dispersion profiles, rotation, and anisotropy.
- In practice, one must also account for systematics, crowding, and membership selection.

Summary of Today's Lecture

- Star clusters evolve under the combined influence of their **internal dynamics** (two-body relaxation, scattering, evaporation, mass segregation, core collapse).
- Their evolution is strongly shaped by the **external galactic potential**: disc, bulge, halo, bar, and dark-matter distribution.
- We explored **orbital dynamics** of clusters in galaxies: conserved quantities, orbital families, axisymmetric vs. barred potentials, Jacobi energy, tidal shocks, and orbital decay.
- The **Jeans equations**, CBE, and Poisson equation provide the theoretical backbone for understanding their kinematics and structural evolution.
- Numerical tools (leapfrog integrators, N-body, `galpy`, `AGAMA`, `Delorean`) allow us to model cluster orbits, tidal fields, and long-term survival.

Key Takeaways

- **Star clusters are dynamical laboratories**: their structure and evolution trace the physics of relaxation, mass segregation, and tidal stripping.
- A cluster's fate is governed by its **orbit in the host galaxy**: eccentricity, pericenter, inclination, bar resonances, and disc crossings determine the strength of tidal forces.
- **Galactic tides** sculpt clusters: tidal truncation, shocks, heating, and the formation of streams and extra-tidal stars.
- Understanding cluster orbits is essential to reconstruct their **formation sites**, migration within the galaxy, and eventual dissolution.
- With modern surveys (Gaia, HST, JWST, ALMA), we can now combine 6D data, orbits, and dynamical modeling to uncover the **life cycle of star clusters** in the Milky Way and beyond.

Star Clusters = Internal Physics + Galactic Environment

Thank You Very Much!

Galactic Dynamics & Star Clusters

An Overview Lecture for MSc and PhD Students

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